UNIT - I

Introduction, Design for Static and Dynamic Loads

1. Mechanical Engineering Design

Design = creative decision-making process that converts an idea / need into a machine element or system which performs the required function safely, reliably and economically.

Machine member = any part (shaft, bolt, gear, spring, bearing ...) whose size and shape are decided by design.

Types of design

- Adaptive design - small changes in an existing design (most common).
- Development design - new design developed from an existing one, existing principle retained.
- New design - entirely new, needs original research (rarest).
- Also: rational design (stress analysis based), empirical design (formulae from experience), industrial design (looks, ergonomics, aesthetics), optimum design (best of all alternatives).

Design process (steps)

Recognition of need → Definition of problem → Synthesis
→ Analysis & optimisation → Evaluation (prototype / test)
→ Presentation (drawings, process sheets, reports)

- *Recognition of need*: a need / defect is felt (low efficiency, cost, noise, safety).
- *Definition*: specify input, output, speed, load, life, space, cost, environment.
- *Synthesis*: think of several possible mechanisms / schemes; select the best few.
- *Analysis*: forces, stresses, deflections -> size; iterate (trial & error) till requirement met.
- *Evaluation*: prototype testing, compare with specification; modify if needed.
- *Presentation*: working drawings with tolerances, material, heat treatment, assembly drawing, BOM.

2. Design Considerations

Factors a designer has to keep in mind while fixing the size and material of a member:

Group	Considerations
Strength / safety	type of load, stress, FOS, stiffness, deflection, vibration
Manufacture	casting / forging / machining, tolerance, surface finish, standard parts
Economy	material cost, labour, tooling, quantity, maintenance cost
Service	wear, corrosion, temperature, lubrication, life, reliability
Human / other	ergonomics, noise, safety guards, aesthetics, weight, space

Factor of safety (n or FOS) = failure stress / allowable (working) stress. For ductile material failure stress = S_y; for brittle = S_{ut}.

$$n = S_y / \sigma_{allow} \quad \text{(ductile)}$$
$$n = S_{ut} / \sigma_{allow} \quad \text{(brittle)}$$

(basic definition)

- Typical n: dead load steady 1.5-2 (ductile) ; live load 2-3 ; shock load 3-5 (up to 8-10 brittle).
- Too high n -> heavy & costly ; too low n -> risk of failure. Depends on: accuracy of load estimate, material scatter, consequence of failure, workmanship.

Note: Stiffness-governed members (machine tool bed, long shafts) are designed for deflection, not for strength alone. Check both.

3. Codes, Standards & Designation of Materials

Standard = set of specifications for size, material, quality, test, terminology, so that parts are interchangeable and safe. Code = set of rules for design / fabrication of a particular item (boiler code, pressure vessel code, AGMA for gears).

- Standards bodies: BIS (IS), ISO, ASME / ANSI, DIN, BS, AGMA, SAE, ASTM, JIS.
- Uses: interchangeability, reduction in variety of sizes, lower cost, better quality, safety, easy replacement, international trade.
- Preferred numbers (R5, R10, R20, R40 series - Renard) are used for standard sizes / speeds: ratio = 5-th root of 10 etc. (R10 -> 1.26 , R20 -> 1.12).

Designation of materials (Indian Standard, IS 1762 / IS 1570)

Material	Designation rule	Example & meaning
Grey cast iron	FG + min. tensile (N/mm^2)	FG 200 : S_{ut} = 200 MPa
Steel (by strength)	Fe + min. UTS	Fe 410 : S_{ut} = 410 MPa
Steel (by yield)	Fe E + min. yield	Fe E 250 : S_y = 250 MPa
Plain carbon steel	digits = C% × 100, C, Mn% × 100	45C8 : 0.45% C , 0.8% Mn
Alloy steel	C%×100, symbols, %×factor	40Ni14 : 0.4% C , 3.5% Ni
Alloy steel (2)	factor 4 : Cr,Ni,Mn,Si,W,Co	35Mn2Mo28 : .35%C, .5%Mn, .28%Mo
Malleable C.I.	BM / WM + UTS	BM 300 : black heart, 300 MPa
Cast steel	CS + UTS	CS 540 : S_{ut} = 540 MPa
Non-ferrous	chemical symbol + %	Cu Zn 40 : brass (60% Cu , 40% Zn)

Rule: Alloy steel: multiplying factor = 4 for Cr, Co, Ni, Mn, Si, W ; 10 for Al, Be, V, Pb, Cu, Nb, Ti, Ta, Zr ; 100 for P, S, N, C (in small amounts).

4. Selection of Materials

Material selection is balanced between properties required, processing and cost. Main criteria:

- Mechanical: strength (S_y, S_{ut}), ductility, hardness, toughness (impact), fatigue strength, creep strength, stiffness E.
- Physical: density (weight), thermal conductivity & expansion, melting point, electrical property.
- Manufacturing: castability, forgeability, machinability, weldability, heat-treatability.
- Service: corrosion / wear resistance, temperature, lubrication, noise damping (grey C.I. best damping).
- Economic: cost per kg, availability, standard sections, scrap value.

Member	Usual material	Reason
Shafts, axles	40C8, 45C8, 40Cr1	strength + toughness
Gears (heavy)	case-hardened 20MnCr5 , 40Ni14	hard teeth, tough core
Machine bed / housing	FG 200 , FG 260	castable, damping, cheap
Springs	55Si2Mn90 , music wire	high elastic limit
Bearing bushes	Bronze, babbitt	low friction, embeddability
Light parts	Al alloys	high strength / weight

5. Design for Static Loads

Static load = load that does not change with time (or changes very slowly). Member is designed so that the induced stress never exceeds the allowable stress.

Modes of failure

- Yielding (ductile) - permanent deformation makes the member unfit : S_y governs.
- Fracture (brittle) - sudden separation : S_{ut} governs ; also stress-raiser cracks.
- Excessive deflection - bad alignment, loss of accuracy (stiffness failure).
- Buckling - slender column / thin shell under compression (Euler).
- Fatigue - under repeated loads at stress < S_y ; creep - slow strain at high temperature ; wear, corrosion, fretting, impact, thermal shock.

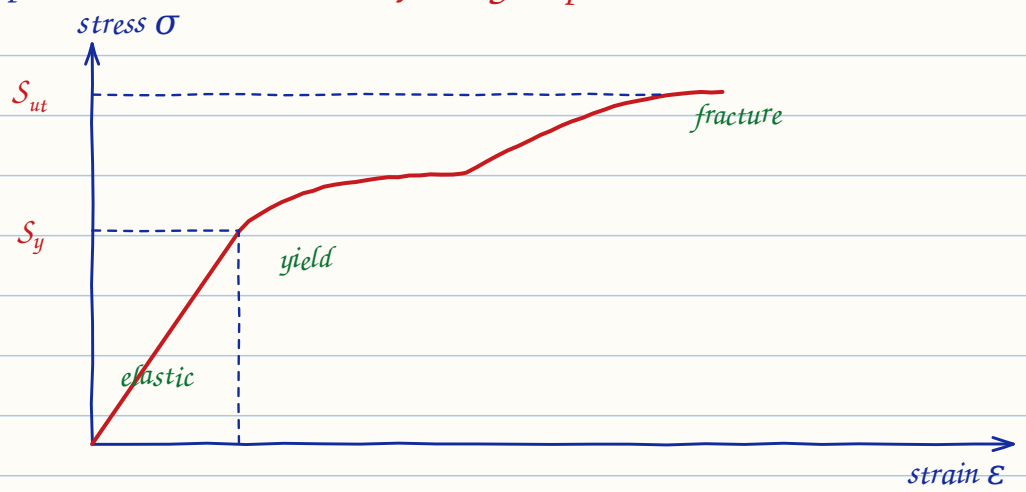


Fig 1: Stress-strain diagram of mild steel (ductile)

Ductile material (elongation > 5%) fails by yielding - design on S_y. Brittle material (C.I., ceramics) fails with no yield - design on S_{ut} (compression strength S_{uc} is much higher: $S_{uc} \approx 3\text{-}4\ S_{ut}$).

(a) Axial loading (tension / compression)

$$\sigma = P / A \le S_y / n$$

$$\text{solid round bar : } d = \sqrt{(4P / \pi \sigma_{allow})}$$

For a long column under compression also check buckling: $P_{cr} = \pi^2 E I / (K L)^2$. Direct shear (pins, keys, rivets): $\tau = P / A_{shear}$, $\tau_{allow} \approx S_{sy}/n$; $S_{sy} = 0.577\, S_y$ (distortion energy) or $0.5\, S_y$ (max shear).

(b) Bending

$$\sigma_b = M y / I = M / Z$$

$$\text{round : } \sigma_b = 32\, M / \pi d^3$$

$$\text{rectangle } (b \times h) : \sigma_b = 6\, M / b h^2$$

Transverse shear stress in beam (max): $\tau_{max} = 1.5\, V/A$ (rectangle), $4V/3A$ (solid circle), at neutral axis. Normally small compared with bending stress - check only for short, deep beams.

(c) Torsion

$$\tau = T r / J$$

$$\text{solid shaft : } \tau = 16\, T / \pi d^3$$

$$\text{hollow } (k = d_i/d_o) : \tau = 16\, T / \pi d_o^3 (1 - k^4)$$

Angle of twist $\theta = T L / (G J)$ rad ; torsional rigidity $G J$. Power : T (N·m) $= 60\, P / (2\pi N) = 9550\, P(kW) / N(rpm)$. Allowable $\tau = S_{sy}/n$; for shafts with keyway reduce by ~25%.

(d) Combined bending + torsion (shafts)

Principal stresses at the surface point where $\sigma = 32M/\pi d^3$ and $\tau = 16T/\pi d^3$ act:

$$\sigma_{1,2} = \sigma/2 \pm \sqrt{((\sigma/2)^2 + \tau^2)}$$

$$\tau_{max} = \sqrt{((\sigma/2)^2 + \tau^2)} = 16 / \pi\, d^3 \cdot \sqrt{(M^2 + T^2)}$$

Define the equivalent twisting moment $T_e = \sqrt{(M^2 + T^2)}$ (Guest - ductile) and equivalent bending moment $M_e = \tfrac{1}{2} [M + \sqrt{(M^2 + T^2)}]$ (Rankine - brittle).

$$d^3 = 16\,T_e / \pi\,\tau_{allow}$$
$$d^3 = 32\,M_e / \pi\,\sigma_{allow}$$

(shaft diameter)

Example 1: Shaft carries M = 1000 N·m, T = 600 N·m. S_y = 300 MPa, n = 2. Find d.

$\tau_{allow} = S_y/2n = 300/4 = 75$ MPa (max shear theory). $T_e = \sqrt{(1000^2 + 600^2)} = 1166$ N·m.

$d^3 = 16 \times 1166\times10^3 / (\pi \times 75) = 79.2\times10^3$ mm³ → $d = 42.9$ mm → *provide d = 45 mm (standard).*

6. Impact (Shock) Loads

A load applied suddenly or falling from a height produces much larger stress than the same load applied gradually. Energy balance : work done by falling load = strain energy stored (½ $P_i\,\delta$).

Case	Equivalent static factor	Stress
Gradual load P	1	$\sigma_{st} = P/A$
Suddenly applied (h = 0)	2	$\sigma = 2\,P/A$
Falling weight W through h	$1 + \sqrt{(1 + 2h/\delta_{st})}$	$\sigma = \sigma_{st}(1 + \sqrt{(1 + 2h/\delta_{st})})$
Moving mass, velocity v	-	$\sigma = \sqrt{(\,m\,v^2\,E\,/\,A\,L\,)}$

$\delta_{st} = W\,L\,/\,A\,E$ (static deflection under W). For large h : $\sigma \approx \sigma_{st}\sqrt{(2h/\delta_{st})} = \sqrt{(2\,W\,h\,E\,/\,A\,L)}$.

- Impact stress can be reduced by increasing L (long bolt), reducing A in unthreaded parts (reduced shank), using springs / washers - all increase the strain energy absorbed.

Example 2: W = 1 kN falls h = 50 mm on a collar of a bar L = 1 m, d = 20 mm, E = 200 GPa. A = 314 mm², $\delta_{st} = 1000\times1000/(314\times200000) = 0.0159$ mm, $\sigma_{st} = 3.18$ MPa.

$\sigma = 3.18\,[\,1 + \sqrt{(1 + 2\times50/0.0159)}\,] = 3.18\,(1 + 79.3) \approx 255$ MPa (80 times static !).

7. Theories of Failure (static loads)

A theory predicts failure under combined (biaxial / triaxial) stress from the simple tension test value. Principal stresses $\sigma_1 \geq \sigma_2 \geq \sigma_3$; for plane stress:

$$\sigma_{1,2} = (\sigma_x+\sigma_y)/2 \pm \sqrt{\left((\sigma_x-\sigma_y)/2\right)^2 + \tau_{xy}^2}$$

Theory (name)	Failure criterion	Design equation	Best for
Max principal stress (Rankine)	$\sigma_1 = S_y$ or S_{ut}	$\sigma_1 \le S/n$	brittle
Max shear stress (Tresca / Guest)	$\tau_{max} = S_y/2$	$\sigma_1 - \sigma_3 \le S_y/n$	ductile (conservative)
Max principal strain (St. Venant)	$\varepsilon_1 = S_y/E$	$\sigma_1 - \mu\sigma_2 - \mu\sigma_3 \le S_y/n$	(rarely used)
Max strain energy (Haigh)	$U = S_y^2/2E$	$\sigma_1^2+\sigma_2^2-2\mu\sigma_1\sigma_2 \le (S_y/n)^2$	(rarely used)
Max distortion energy (von Mises-Hencky)	$\sigma' = S_y$	$\sigma' = \sqrt{(\sigma_1^2-\sigma_1\sigma_2+\sigma_2^2)} \le S_y/n$	ductile (best fit)

$$\text{von Mises : } \sigma' = \sqrt{\sigma_x^2 - \sigma_x\sigma_y + \sigma_y^2 + 3\tau_{xy}^2}$$
$$\text{bending + torsion : } \sigma' = \sqrt{\sigma^2 + 3\tau^2}$$

(DET)

- Pure shear, ductile : Tresca $S_{sy} = 0.5\ S_y$; von Mises $S_{sy} = 0.577\ S_y$ (experiments agree with 0.577).
- Tresca is safer (hexagon inside the von Mises ellipse) ; difference at most 15%.
- Brittle material : use Rankine (or Coulomb-Mohr when $S_{uc} \neq S_{ut}$) : $\sigma_1/S_{ut} - \sigma_3/S_{uc} = 1/n$.

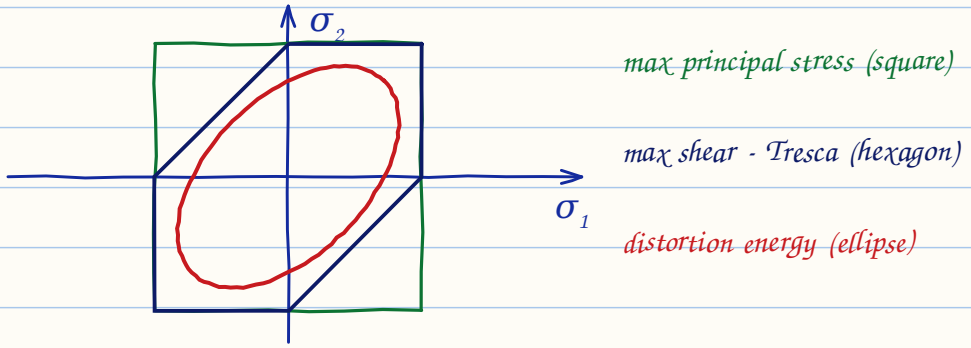


Fig 2: Failure loci in the σ_1 - σ_2 plane

8. Design for Dynamic (Fatigue) Loads

Fatigue : failure of a member under repeated / fluctuating stress at a value well below S_{ut} (even below S_y), after a number of cycles N. About 90% of service failures. Starts as micro-crack at a stress raiser → propagates (beach marks) → sudden final fracture of the remaining area.

Endurance limit & S-N curve (R. R. Moore test)

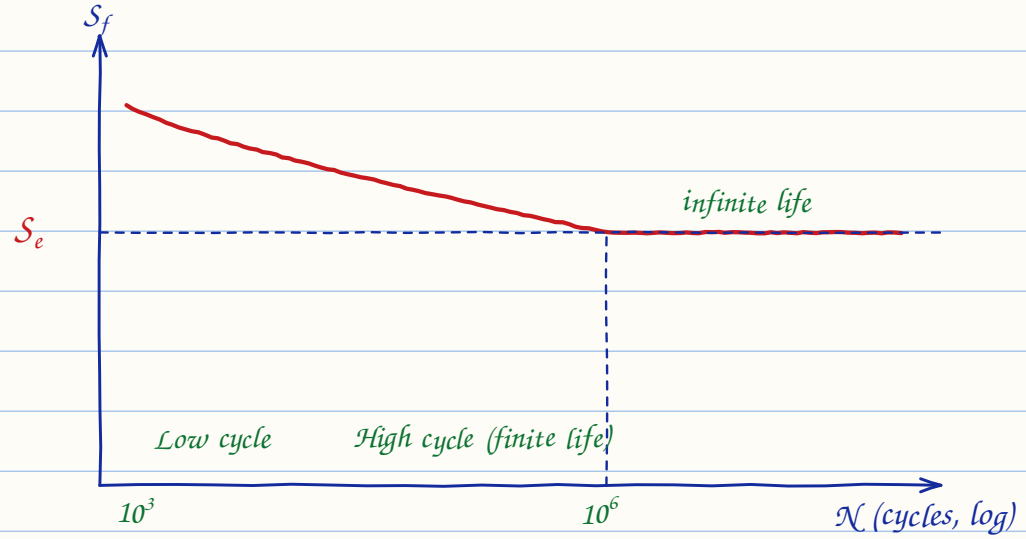


Fig 3: S-N curve for steel (knee at 10^6 cycles)

- Endurance limit S_e' = max completely reversed stress for infinite life ($N > 10^6$) of a standard polished specimen (7.5 mm dia).
- Steels show a knee ; non-ferrous (Al, Cu) have no knee - fatigue strength quoted at 10^8-5×10^8 cycles.

$$S_e' = 0.5\, S_{ut} \quad (S_{ut} \le 1400\ MPa)$$
$$S_e' = 700\ MPa \quad (S_{ut} > 1400\ MPa)$$

(steel, bending)

Also : cast iron $S_e' \approx 0.4\, S_{ut}$; Al alloys $S_e' \approx 0.4\, S_{ut}$ (≤ 130 MPa). Fatigue strength at 10^3 cycles $\approx 0.9\, S_{ut}$ (bending), $0.75\, S_{ut}$ (axial). Between 10^3 and 10^6: log S_f varies linearly with log N (Basquin), $S_f = a\, N^b$.

Modified (corrected) endurance limit - Marin factors

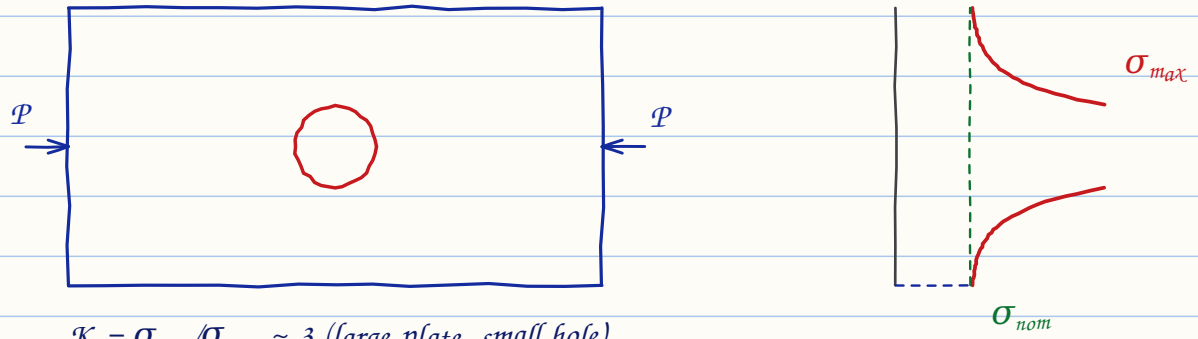
$$S_e = k_a\, k_b\, k_c\, k_d\, k_e \cdot S_e'$$

Factor	Meaning	Typical value
k_a	surface finish (mirror 1.0 ; ground 0.9 ; machined 0.78 ; hot rolled 0.6 ; forged 0.5)	0.5 - 1
k_b	size : 1.24 $d^{-0.107}$ (2.79-51 mm) ; axial load → 1 ; Bhandari : 1 (d<7.5), 0.85 (7.5-50), 0.75 (>50)	0.7 - 1
k_c	load type : bending 1.0 ; axial 0.85 ; torsion 0.59	0.59 - 1
k_d	temperature : 1 at room T, falls above 450°C	≤ 1
k_e	reliability : 50% 1.0 ; 90% 0.897 ; 95% 0.868 ; 99% 0.814 ; 99.9% 0.753	0.75 - 1

For torsion (ductile) use $S_{se} = 0.577\ S_e$ (or simply apply $k_c = 0.59$ to S_e'). Shear yield $S_{sy} = 0.577\ S_y$.

9. Stress Concentration & Notch Sensitivity

Holes, shoulders, keyways, grooves, threads, sharp corners → abrupt change of section → local stress much higher than nominal (flow of force lines crowd together).



$K_t = \sigma_{max}/\sigma_{nom} \approx 3$ (large plate, small hole)

Fig 4: Stress concentration at a hole

$$K_t = \sigma_{max} / \sigma_{nom} \quad \text{(theoretical, from geometry only)}$$
$$K_f = 1 + q\,(K_t - 1) \quad \text{(fatigue stress conc. factor)}$$
$$q = (K_f - 1)/(K_t - 1) \quad , \quad 0 \le q \le 1$$

- Notch sensitivity q : $q = 0$ → material insensitive (K_f=1) ; $q = 1$ → fully sensitive (K_f=K_t). Depends on material and notch radius r : larger r , higher S_{ut} → larger q. Cast iron $q \approx 0.2$ (internal flaws) ; hardened steel $q \to 1$.
- Static loads : ductile material - K_t is ignored (local yielding relieves it) ; brittle material - K_t must be applied. Fatigue : K_f is always applied (on alternating stress).

Methods of reducing stress concentration

- Generous fillet radius at shoulders ; relief grooves ; undercuts ; gradual taper.
- Additional (stress-relief) notches / holes on the shoulder that make force flow smooth ; avoid sharp keyways at critical sections - use profile (sled-runner) keyways.
- Cold working, shot peening, surface rolling → compressive residual stress, improves fatigue strength.

10. Types of Fluctuating Loads

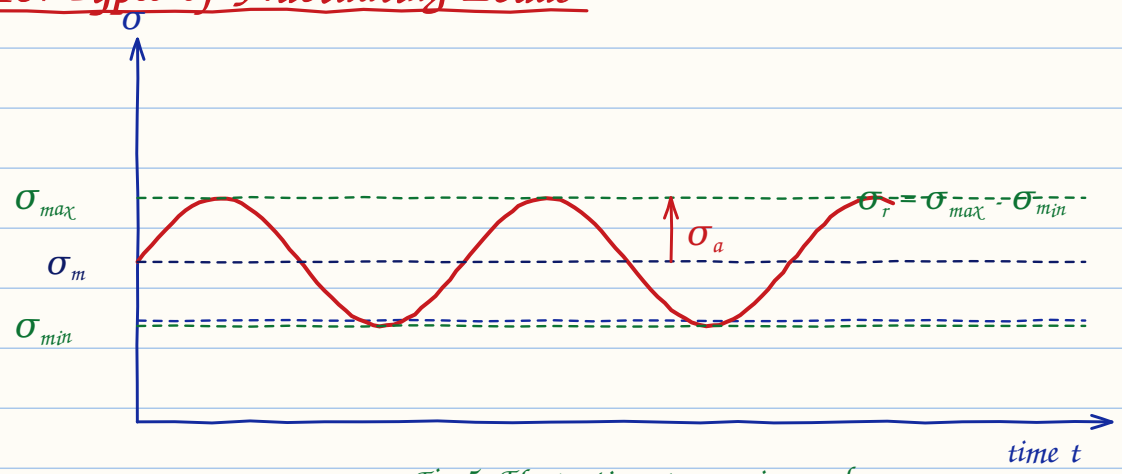


Fig 5: Fluctuating stress - time cycle

$$\sigma_m = (\sigma_{max} + \sigma_{min})/2 \quad , \quad \sigma_a = (\sigma_{max} - \sigma_{min})/2$$

$$\mathcal{R} = \sigma_{min}/\sigma_{max} \quad , \quad \mathcal{A} = \sigma_a/\sigma_m$$

Type	σ_m	σ_a	$\mathcal{R}$
Completely reversed (rotating shaft)	0	σ_{max}	-1
Repeated / pulsating (gear tooth)	$\sigma_{max}/2$	$\sigma_{max}/2$	0
General fluctuating (spring, bolt)	$\neq 0$	$\neq 0$	$-1 < \mathcal{R} < 1$
Steady (static)	σ	0	+1

11. Fatigue Design for Infinite Life

For a member under σ_m and σ_a the (σ_m, σ_a) point must lie below the failure line of the Haigh / Goodman diagram. Endpoints : $\sigma_a = S_e$ on the vertical axis ($\sigma_m = 0$), $\sigma_m = S_{ut}$ or S_y on the horizontal axis.

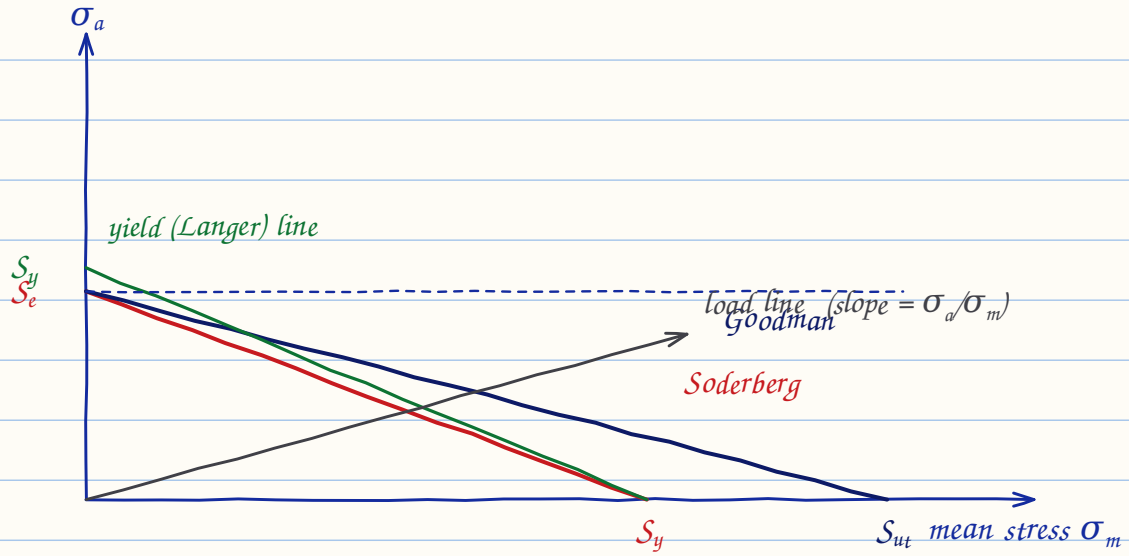


Fig 6: Failure criteria on the σ_m - σ_a diagram

Criteria (n = factor of safety)

Soderberg : $\sigma_a/S_e + \sigma_m/S_y = 1/n$

Goodman : $\sigma_a/S_e + \sigma_m/S_{ut} = 1/n$

Gerber : $n\,\sigma_a/S_e + (n\,\sigma_m/S_{ut})^2 = 1$

- Soderberg - most conservative (line to S_y ; also guards against yielding) ; used when safety is vital.
- Goodman - fits brittle-material data better ; straight line S_e to S_{ut}.
- Modified Goodman - Goodman line + a yield (Langer) line $\sigma_a + \sigma_m = S_y$ to check first-cycle yielding. Design point must lie inside both.
- Gerber parabola - best fit to ductile test data (average, less safe) ; Gerber / Goodman are used for reliable loads.
- Stress concentration : use K_f on the alternating part ($\sigma_a = K_f\,\sigma_{a,nom}$) ; mean stress normally not multiplied for ductile steels ($K_{fm} = 1$) but is for brittle ones.

Remember: Use S_e (corrected, with k_a...k_e) and K_f in every fatigue equation. In axial / torsion cases replace S_e, S_y, S_{ut} by S_{se}, S_{sy}, $S_{su} = 0.577\,S_y$, $0.8\,S_{ut}$.

Example 3: A steel bar (S_{ut} = 600, S_y = 380 MPa) has S_e = 220 MPa (all corrections) and K_f = 1.5. Axial load varies 10 kN to 30 kN. Find d for n = 2 by Soderberg.

P_m = 20 kN , P_a = 10 kN. $\sigma_m = 20000/A$, $\sigma_a = 1.5 \times 10000/A$.

$1/2 = (15000/A)/220 + (20000/A)/380 = (68.2 + 52.6)/A \rightarrow A = 241.6\ mm^2 \rightarrow d = \sqrt{(4A/\pi)} = 17.5 \rightarrow d = 18\ mm.$

(Goodman with the same data : $A = 203\ mm^2 \rightarrow d = 16.1\ mm$, i.e. less conservative.)

12. Fatigue Design under Combined Stresses

Shafts carry bending (σ) and torsion (τ) that are each fluctuating : σ_m, σ_a, τ_m, τ_a. Use the distortion-energy theory to convert them to an equivalent von Mises mean and alternating stress, then apply Soderberg / Goodman.

$$\sigma_a' = \sqrt{\left((K_f \sigma_a)^2 + 3\,(K_{fs} \tau_a)^2 \right)}$$
$$\sigma_m' = \sqrt{\left(\sigma_m^2 + 3\,\tau_m^2 \right)}$$

$$1/n = \sigma_a'/S_e + \sigma_m'/S_y \quad \text{(Soderberg)}$$
$$1/n = \sigma_a'/S_e + \sigma_m'/S_{ut} \quad \text{(Goodman)}$$

- S_e here is the (bending) endurance limit with k_a, k_b, k_d, k_e ; the torsion effect $k_e = 0.59$ is already taken by the $3\,\tau_a^2$ term - do not apply it twice.
- Rotating shaft, steady torque T and bending M : $\sigma_a = 32\,M/\pi d^3$, $\sigma_m = 0$, $\tau_m = 16\,T/\pi d^3$, $\tau_a = 0 \rightarrow \sigma_a' = K_f\,32M/\pi d^3$, $\sigma_m' = \sqrt{3} \cdot 16T/\pi d^3$.

$$(32\,n / \pi\,d^3) \cdot \sqrt{\left((K_f\,M / S_e)^2 + (3/4)(T / S_y)^2 \right)} = 1 \quad \text{(Soderberg - shaft)}$$

- Steps : (1) loads & reactions → M, T diagrams ; (2) critical section ; (3) choose material → S_{ut}, S_y ; (4) S_e', Marin factors → S_e ; (5) K_t, q → K_f ; (6) assume d, find σ_a, σ_m ; (7) apply criterion → d ; (8) round to standard size & check.

Finite life (quick note)

If life N is between 10^3 and 10^6 : $S_f = a\,N^b$, $a = (f\,S_{ut})^2/S_e$, $b = -(1/3)\log_{10}(f\,S_{ut}/S_e)$, $f \approx 0.9$; use S_f instead of S_e in the criterion. Cumulative damage (Miner) : $\Sigma\ n_i/N_i = 1$.

Quick revision - Unit I

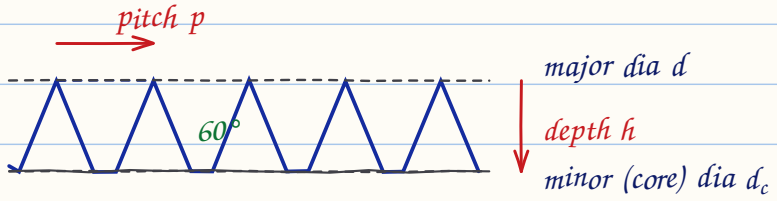
Topic	Key formula / idea
Design stress	$\sigma_{allow} = S_y/n$ (ductile) ; S_{ut}/n (brittle)
Shaft (static)	$d^3 = 16\sqrt{(M^2+T^2)} / (\pi\ \tau_{allow})$
Impact	$\sigma = \sigma_{st}[1 + \sqrt{(1 + 2h/\delta_{st})}]$
von Mises	$\sigma' = \sqrt{(\sigma^2 + 3\tau^2)} \leq S_y/n$
Endurance limit	$S_e = k_a k_b k_c k_d k_e (0.5\ S_{ut})$
Notch	$K_f = 1 + q(K_t - 1)$
Fatigue (infinite life)	$\sigma_a/S_e + \sigma_m/S_{y\ or\ ut} = 1/n$

UNIT - II

Design of Bolted and Welded Joints

1. Threaded Fasteners

Detachable joints - bolts, screws, studs, nuts - clamp parts together by tension in the shank created by tightening the nut. Threads are an inclined plane (helix) wrapped on a cylinder.



ISO metric thread - V form, 60°

Fig 1: Screw thread terminology

- Pitch p : axial distance between adjacent threads ; lead $L = n\ p$ (n = number of starts) ; single start → lead = pitch.
- Major dia d (nominal) , minor / core dia d_c , pitch dia $d_p = (d + d_c)/2$; helix angle α : $\tan \alpha = L / \pi d_p$.
- Tensile stress area $A_t = (\pi/4)(d - 0.9382\ p)^2 \approx (\pi/4)\ d_c^2$ with $d_c \approx 0.8\ d$. Always use A_t (or core area) for stress in the bolt.
- Metric designation : M20 × 2.5 → nominal dia 20 mm, coarse pitch 2.5 ; M20 × 1.5 → fine thread (better for vibration, shallow depth, bigger core dia, adjustment).

Thread form	Features	Use
ISO metric (V, 60°)	high friction, strong core, self-locking	bolts, nuts, screws
Square	efficient, no radial thrust, difficult to cut	power screws, jacks
Acme (29°)	easy to make, wear adjustable by split nut	lead screws, vices
Buttress	strong in one direction only	press, jack, vice
Knuckle / round	rounded, resists damage, dirt	bulbs, hose couplings

Common fasteners & locking devices

- Through bolt (nut at the other end) , tap bolt (screws into a tapped hole, nut not used) , stud (threaded both ends, for cylinder heads) , cap screw , set screw (hold hub on shaft).
- Locking devices against vibration loosening : jam (lock) nut, castle nut + split pin, spring (Grover) washer, toothed lock washer, Sarcotite / nylon insert (Nylock) nut, wire locking, tab washer, set screw.

Bolt grades (property class)

Class a.b : S_{ut} = 100 a MPa ; S_y = (b/10) S_{ut}.

Class	4.6	5.8	8.8	10.9	12.9
S_{ut} (MPa)	400	500	800	1000	1200
S_y (MPa)	240	400	640	900	1080

2. Preload & Stresses in Bolts

Bolts are always tightened to an initial tension (preload) F_i. Preload keeps the joint compressed, so that an external load does not open it, increases fatigue life (bolt sees only a part of the load), and prevents leakage / loosening.

$$F_i = 2840\ d\ N \quad (d \text{ in mm ; Bhandari - reusable joints})$$

$$F_i = 0.75\ A_t\ S_p \text{ (reusable)} , \quad 0.90\ A_t\ S_p \text{ (permanent)}$$

S_p = proof strength ≈ 0.85 S_y (class 8.8 : 600 MPa ; 10.9 : 830 MPa).

Various stresses induced in bolts

Cause	Stress	Formula
Initial tightening	direct tension	$\sigma_t = F_i / A_t$
Torque while tightening	torsional shear	$\tau = 16\ T_1 / \pi\ d_c^3$ ($T_1 \approx$ ½ T)
External axial load P	additional tension	$\sigma = (F_i + C\ P)/A_t$
Transverse (shear) load	direct shear	$\tau = F_s/(\pi/4\ d^2)$, bearing $\sigma_c = F_s/(d\ t)$
Thread contact	shear of threads	$\tau = F / (\pi\ d_c \cdot 0.5\ p \cdot z)$; nut H ≈ 0.8d safe
Non-parallel seats	bending	$\sigma_b = E\ d\ \theta\ /\ 2L$ - avoid by spot facing

Note: Because tightening twists the bolt, design on the combined stress : bolt is made ~30% stronger : $A_t \geq 1.3\ F_i / \sigma_{allow}$ (or use von Mises with σ, τ from torque).

Design of a bolt for static loads

- Axial pull P : $A_t = P / \sigma_{allow}$, $\sigma_{allow} = S_y/n$ (n = 2-3 for static) → choose standard M-size from the table of core areas (M10 : 58 mm² , M12 : 84 , M16 : 157 , M20 : 245 , M24 : 353 mm²).
- Transverse (shear) load : turned / fitted bolt in reamed hole : $\tau = 4\ F_s / (\pi\ d^2) \leq S_{sy}/n$; crushing $\sigma_c = F_s/(d\ t)$. Ordinary bolt in clearance hole : load is carried by friction, $\mu\ F_i\ i \geq F_s$ ($\mu \approx 0.15$-0.2, i = number of interfaces) → large preload needed.
- Bolts of uniform strength : to improve shock / fatigue strength drill axial hole in the shank or reduce shank dia to core dia (d_c) so that the strain energy stored ($\frac{1}{2}\ \sigma^2\ A\ L / E$) is spread evenly and thread-root stress concentration is not the weak point.
- Nut height : $H = 0.8\ d$ standard ; thread-stripping shear area $\approx \pi\ d_c\ (0.5\ H)$ per nut ; nut is made softer (grade ≈ one lower than bolt) so load spreads over the first 3 threads - first thread carries ~ 30%, 3rd ~ 10%. Using a taller nut does not help beyond ~ 6 threads.

Example (axial) : bolt carries a steady pull of 20 kN, class 4.6 ($S_y = 240$), $n = 3 \rightarrow \sigma_{allow} = 80$ MPa. $A_t = 20000/80 = 250$ mm² → M20 ($A_t = 245 \approx$ OK) ; with torque factor 1.3 → 325 mm² → M24.

Torque requirement for tightening

$$T = F_i \left[(d_p/2)\left((\mu + \tan\alpha)/(1 - \mu\ \tan\alpha) \right) + \mu_c\ d_c/2 \right]$$

$$T = K\ F_i\ d \quad (K \approx 0.2 \text{ for dry / plain steel})$$

- First term : thread (screw) friction ; second : collar / nut face friction (mean collar dia d_c). K (torque coefficient) = 0.15 lubricated , 0.2 dry , 0.3 rough / plated.
- Efficiency of tightening is poor : ~ 10% of torque makes tension, rest is lost in friction → large scatter ±25% unless torque wrench or turn-of-nut method is used.

- Spanner length for hand tightening ≈ 15-20 d. Standard spanner on M16 → easily overloads M8 - avoid.

Screw mechanics - self-locking & efficiency

$$\tan \alpha = p / \pi d_p \quad , \quad \mu' = \mu / \cos 30° \ (V \text{ thread})$$

Self-locking if $\mu' > \tan \alpha$ (always true for fasteners, $\alpha \approx 2\text{-}3°$)

$$\eta = \tan \alpha \ (1 - \mu' \tan \alpha) / (\mu' + \tan \alpha) \approx 20\text{-}25\ \%$$

M20 × 2.5 : $d_p = 18.4$ mm , $\tan \alpha = 0.0433$; with $\mu' = 0.15 \rightarrow \eta = 22$ %. Low efficiency = good locking but 78% of the torque goes in friction ; lubrication changes F_i by ±30% for the same torque !

Example 1 : M16 bolt (d = 16 mm). $F_i = 2840 \times 16 = 45.4$ kN, $T = 0.2 \times 45.4 \times 10^3 \times 16 = 145$ N·m. Core area ≈ 144 mm² (M16×2) → $\sigma = 45400/144 = 315$ MPa → class 8.8 or higher required.

3. Bolted Joint with External Load - Joint Stiffness

When an external tensile load P (axial, pulls the plates apart) is applied to a preloaded joint, the bolt stretches more and the compressed plates relax. Bolt stiffness $k_b = A_b E / L$, member (plates + washers) stiffness k_m (conical pressure frustum model).

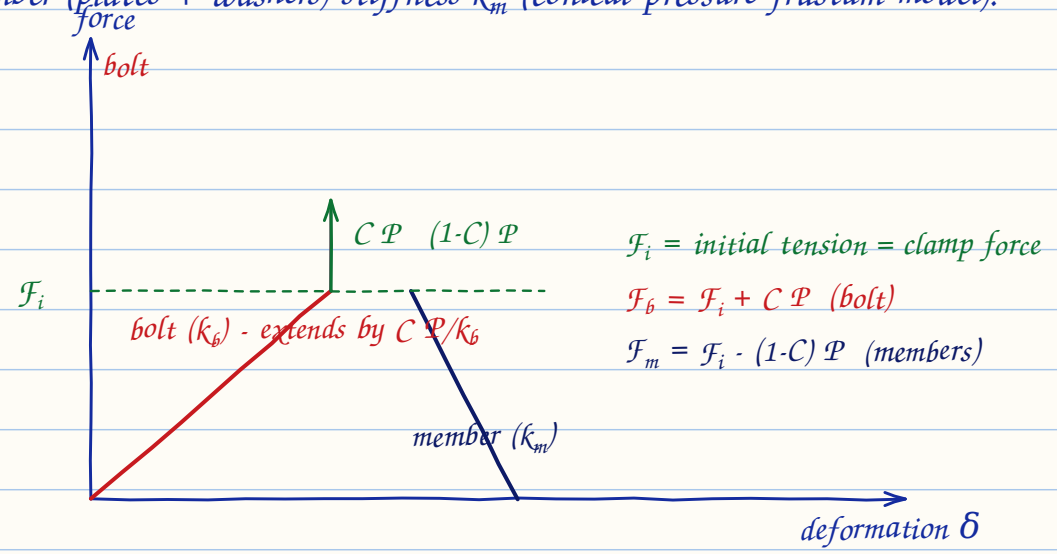


Fig 2: Load - deformation diagram of a preloaded joint

$$C = k_b / (k_b + k_m) \quad \text{(joint constant, 0.1 - 0.4)}$$

$$F_b = F_i + C P \quad , \quad F_m = F_i - (1 - C) P$$

- Separation of the joint (leak) when $F_m = 0 \rightarrow P_o = F_i / (1 - C)$. Design : $P < P_o$; factor of safety against separation $n_o = P_o/P$.
- Smaller C (stiff plates, flexible long bolt) → bolt takes only a small part of P → better fatigue life ; this is why long thin bolts, reduced shank and hard plates are good.
- Bolt must also not yield : $n_p = (S_p A_t - F_i) / C P$ (load factor).

Gasketed joints (pressure vessel covers, cylinder heads)

Total force on the cover $P_{total} = (\pi/4) D^2 p_{fluid}$. With z bolts on bolt-circle diameter D_b : $P = P_{total}/z$ per bolt. A soft gasket has small k_m → $C \approx 0.6$-0.9 (the bolt takes most of P) , a hard / metal gasket → $C \approx 0.2$-0.4.

- Gasket must retain a minimum seating pressure for no leak : $F_m = F_i - (1 - C)P \geq$ gasket seating load. Hence a high F_i ($\approx 0.9 A_t S_p$) is chosen in leak-proof joints.
- Number of bolts & pitch : bolt pitch = $\pi D_b/z$. Keep pitch ≥ 3d (spanner space) and ≤ $20\sqrt{t}$ (t = flange thickness, mm) so that the gasket pressure stays uniform. For high-pressure joints pitch ≈ 3d to 5d ; at least 4 bolts, usually even.

$$\text{Resultant load on a bolt : } P_{res} = P_i + C P \leq \sigma_{allow} A_t$$

Example 2 : cylinder cover, D = 300 mm, p = 1.5 MPa, 8 bolts. $P_{total} = (\pi/4)\, 300^2 \times 1.5 = 106$ kN → P = 13.25 kN per bolt. Take C = 0.3, $F_i = 2840$ d. With M16 : $F_i = 45.4$ kN. $F_b = 45.4 + 0.3 \times 13.25 = 49.4$ kN → $\sigma = 49400/144 = 343$ MPa < $0.85 S_y$ for class 8.8 (544). Safe ; also $F_m = 45.4 - 0.7 \times 13.25 = 36.1$ kN > 0 (no leakage).

Fatigue loading of bolts (fluctuating P)

$$\sigma_a = K_f C (P_{max} - P_{min}) / 2 A_t$$
$$\sigma_m = [\, F_i + C (P_{max} + P_{min})/2 \,] / A_t$$
$$\text{Goodman : } \sigma_a/S_e + \sigma_m/S_{ut} = 1/n$$

K_f (threads) ≈ 2.2 (rolled) to 3.8 (cut, hard) ; S_e for bolts (axial) ≈ $0.5 S_{ut} \times k_a$ … ~ 130-180 MPa for class 8.8 rolled threads. Higher F_i → σ_m higher but σ_a unchanged ; bolt life improved by preload up to proof limit.

Example 3a (fatigue of a cover bolt) : M12 ($A_t = 84$ mm²), class 8.8 ($S_{ut} = 800$), rolled threads $K_f = 3$, $S_e = 150$ MPa. P varies 0 → 6 kN per bolt, $F_i = 15$ kN, C = 0.25. $\sigma_a = 3 \times 0.25 \times 6000 / (2 \times 84) = 26.8$ MPa. $\sigma_m = (15000 + 0.25 \times 3000)/84 = 187.5$ MPa.

Goodman : $1/n = 26.8/150 + 187.5/800 = 0.413 \rightarrow n = 2.4$. Yield check : $F_{b,max} = 15000 + 0.25\times6000 = 16.5$ kN $\rightarrow 196$ MPa $< S_p = 600$ (OK).

Steps in bolted joint design

- (1) Find external load per bolt P (pressure × area / z) ; (2) choose C from gasket / plate stiffness ; (3) choose F_i (2840 d or 0.75-0.9 $A_t S_p$) ; (4) $F_b = F_i + C P \rightarrow A_t = 1.3 F_b / \sigma_{allow}$; (5) check separation $P_o = F_i/(1 - C) > P$; (6) check fatigue (Goodman) if load fluctuates ; (7) select standard size, torque $T = 0.2 F_i d$.

4. Eccentrically Loaded Bolted Joints

(a) Load in the plane perpendicular to bolt axes (bracket in shear)

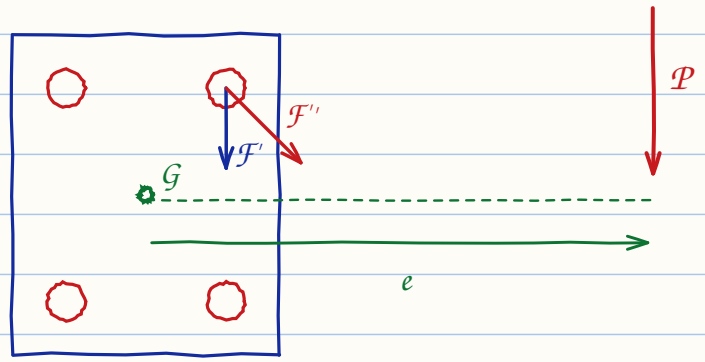


Fig 3: Eccentric load P on a bolt group (bolts in shear)

- Shift P to the centroid G of the bolt group : P (direct) + moment $M = P e$.
- Primary (direct) shear $F' = P / n$ on every bolt, parallel to P.
- Secondary shear $F''_i = M r_i / \Sigma r^2$ (same for equal bolts : $F'' = P e / (n r)$), perpendicular to the line joining the bolt to G.

$$F = \sqrt{(F'^2 + F''^2 + 2 F' F'' \cos\theta)}$$

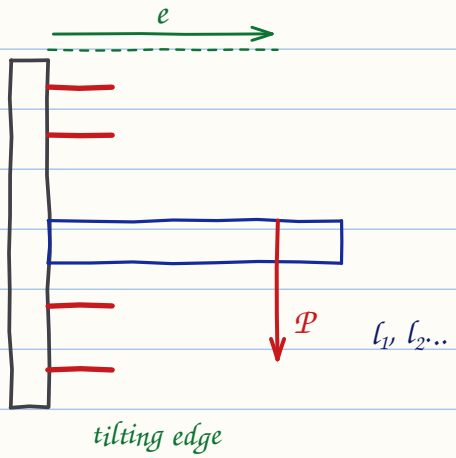
$$F'' \propto r \rightarrow F''_1/r_1 = F''_2/r_2 = \ldots$$

Bolt design : $\tau = F / (\pi/4\, d_c^2) \le S_{sy}/n$ ($S_{sy} = 0.5 S_y$) ; check bearing on the plate $\sigma_c = F/(d t) \le S_y/n$. If bolts are in clearance holes the load is carried by friction $\mu F_i n_s \ge P$ (slip-critical).

Example 3 : 4 bolts at corners of a 100 mm square ($r = 70.7$ mm), $P = 12$ kN, $e = 200$ mm. $F' = 3000$ N ; $F'' = 12000 \times 200 / (4 \times 70.7) = 8485$ N. For the worst bolts $\theta = 45°$: $F = \sqrt{(3000^2 + 8485^2 + 2\times3000\times8485\times0.707)} = 10.8$ kN. Class 4.6 ($S_y = 240$), $n =$

2.5 : $\tau_{allow} = 0.5 \times 240/2.5 = 48$ MPa → $d_c^2 = 4 \times 10820/(\pi \times 48) = 287$ → $d_c = 16.9$ mm → **M20** ($d_c = 17.3$).

(b) Load parallel to the bolt axes, eccentric (bracket in tension)



$F_i \propto$ distance l from the tilting edge

$P\,e = \Sigma\, F_i\, l_i = (F_1/l_1)\, \Sigma\, l_i^2$

Fig 4: Bracket tilting about the lower edge

Under P e the bracket tends to tilt about the edge of the base (assuming the plate rigid, the bolt elongation ∝ distance l from the tilting edge ; bolts near the edge carry little).

$$F_i \text{ (bolt at } l_i) = F_1\, l_i/l_1$$

$$P\,e = \Sigma\, F_i\, l_i \;\rightarrow\; F_1 \text{ (max)} = P\,e\, l_1/\Sigma\, l^2$$

$$\text{Direct load per bolt} = P/z \;\rightarrow\; \text{total } F_{max} = P/z + P\,e\, l_1/\Sigma\, l^2$$

- Initial tightening is superimposed ; if joint-compressive preload is large, the whole base remains in contact (the tilt-edge assumption is conservative). Design : $A_t = F_{max}/\sigma_{allow}$.

Example 6 : bracket with 4 bolts (2 rows × 2) in tension, $P = 20$ kN at $e = 200$ mm from the wall. Lower bolt row is the tilt-edge side : rows are at $l_1 = 150$ mm (upper, farthest) and $l_2 = 50$ mm. Per row 2 bolts. $\Sigma\, l^2 = 2\,(150^2 + 50^2) = 50000$ mm². $F_1 = P\,e\, l_1/\Sigma\, l^2 = 20000 \times 200 \times 150/50000 = 12$ kN per bolt. Direct load $P/4 = 5$ kN → $F_{max} = 17$ kN. Class 5.8 ($S_y = 400$) , $n = 2$ → $\sigma_{allow} = 200$ MPa → $A_t = 17000/200 = 85$ mm² → **M12** (84 mm², with preload fine - take M14).

- Combined case (load inclined / both shear + tension) : σ_t and τ on the same bolt - use equivalent stress $\sigma_{eq} = \sqrt{(\sigma_t^2 + 3\tau^2)}$ or maximum shear theory $\sqrt{(\sigma_t^2 + 4\tau^2)} \le S_y/n$.

5. Welded Joints

Permanent joint - parts fused with / without filler metal (arc, gas, resistance, TIG, MIG, submerged arc). Advantages : no hole weakening, lighter, cheaper, leak-proof, can join complex shapes. Disadvantages : residual stress & distortion, poor fatigue, skilled work needed, heat-affected zone.

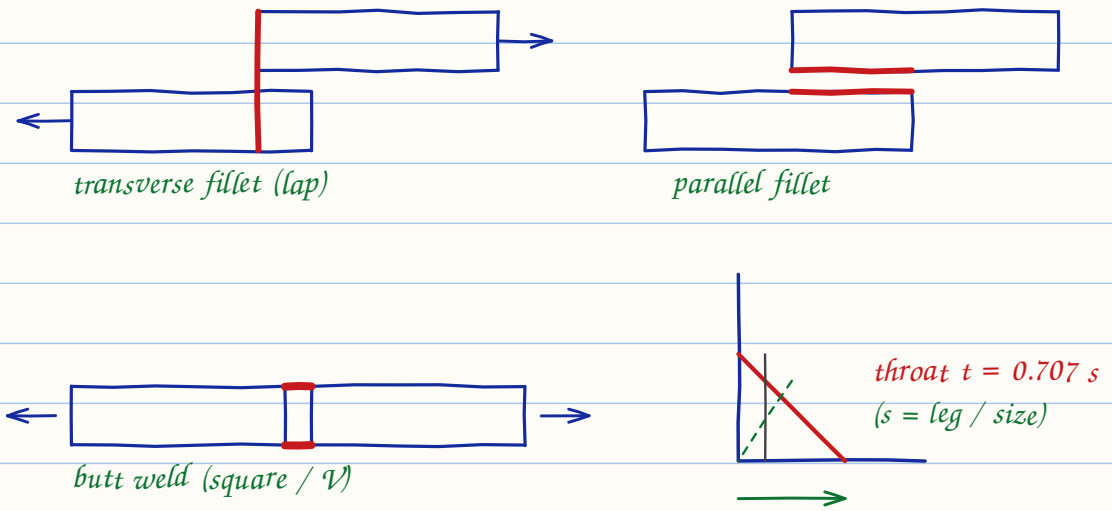


Fig 5: Types of welded joints, fillet throat

- Butt joint (square, single / double V, U, J, bevel) - plates in the same plane ; lap joint (fillet transverse or parallel) ; T joint , corner , edge ; plug / slot / spot welds.
- Design standards : IS 816 (arc welding of mild steel), AWS D1.1, IS 9595. Weld symbol : arrow + reference line + weld symbol (triangle = fillet, V = single V groove) + size s + length.

6. Strength of Butt and Fillet Welds

Butt weld : $P = t\, l\, \sigma_t$

(t = plate thickness = throat , l = length)

Transverse fillet (single) : $P = 0.707\, s\, l\, \sigma_t$

Parallel fillet (single) : $P = 0.707\, s\, l\, \tau$

Double parallel fillet : $P = 1.414\, s\, l\, \tau$

- Fillet weld cross-section is a right triangle of leg s ; failure plane through the throat $t = s \cos 45° = 0.707\, s$. For a transverse fillet the plane is at 67.5° and the stress is ~ τ (conservative to treat it like this).

- Allowable (static, IS 816) : butt in tension 90-110 MPa ; fillet in shear 80 MPa ; reduce by 0.8 (field weld) and for fatigue use K_f below. Weld strength = 1 for reinforced full-penetration butt weld.

Weld location	K_f (fatigue)
Reinforced butt weld	1.2
Toe of transverse fillet weld	1.5
End of parallel fillet weld	2.7
T-butt joint, sharp corner	2.0

- Combined lap joint (transverse l_1 + double parallel l_2) : $P = 0.707\ s\ l_1\ \tau + 1.414\ s\ l_2\ \tau$ (total effective length). Plug / spot weld : $P = (\pi/4)\ d^2\ \tau$, d = hole / spot diameter.
- Joint efficiency η_w (strength of joint / strength of plate) : full-penetration butt 0.9-1.0 ; double fillet lap ≈ 0.7-0.8 ; single fillet ≈ 0.55. Design load = η_w × plate strength.

Axially loaded unsymmetrical welds (angle section on a plate)

Line of action of P passes through the centroid G of the angle ; welds l_1 and l_2 at distances y_1, y_2 from G must have equal moment about G so that no eccentricity arises :

$$l_1\ y_1 = l_2\ y_2 \quad \text{and} \quad l_1 + l_2 = l$$
$$P = 0.707\ s\ (l_1 + l_2)\ \tau_{allow}$$

Example 4 : Plate 100 × 10 mm is welded to a frame by two parallel fillet welds (one each side) to carry $P = 80$ kN. $\tau_{allow} = 80$ MPa. $P = 1.414\ s\ l\ \tau \rightarrow s\ l = 80000 / (1.414 \times 80) = 707$ mm². Take $s = 8$ mm $\rightarrow l = 88.4$ mm ; add 2s for crater / end return → provide $l = 105$ mm on each side ($l \geq$ plate width 100 mm, OK).

7. Welds in Bending and Torsion

(a) Welds subjected to bending

Treat the weld group as a line of unit throat width. Then $\sigma = M / (0.707\ s\ Z_u)$, where Z_u is the section modulus of the weld lines (mm²).

Weld group (lines)	Z_u	J_u (torsion)
one vertical line, length d	$d^2 / 6$	$d^3 / 12$
two vertical lines (d long, b apart)	$d^2 / 3$	$d(3b^2 + d^2)/6$
two horizontal lines (b long, d apart)	$b\,d$	$b(3d^2 + b^2)/6$
rectangle all round ($b \times d$)	$d(3b + d)/3$	$(b + d)^3/6$
circular weld (dia d)	$\pi\, d^2 / 4$	$\pi\, d^3 / 4$

Transverse shear V adds $\tau' = V / (0.707\, s\, L)$, L = total weld length ; resultant $\tau = \sqrt{(\tau'^2 + \tau_b^2)}$.

(b) Eccentric load in the plane of the weld (torsion)

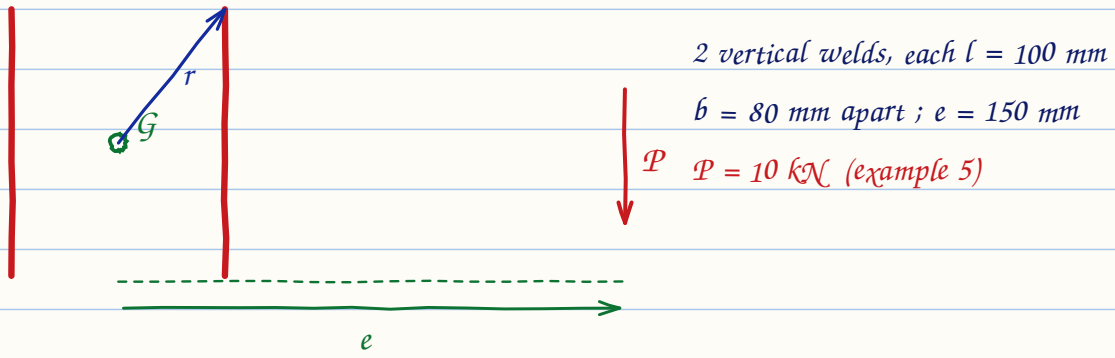


Fig 6: Eccentrically loaded weld group

$$\tau' = P / A = P / (0.707\, s\, L) \quad \text{(primary shear)}$$
$$\tau'' = T\, r / J = P\, e\, r / J \quad , \quad J = 0.707\, s\, J_u$$
$$J_u = \Sigma\, (l_i^3/12 + l_i\, r_i^2) \quad \text{(about } G\text{)}$$
$$\tau = \sqrt{(\tau'^2 + \tau''^2 + 2\, \tau'\, \tau''\, \cos\theta)} \le \tau_{allow}$$

- Locate centroid G : $\bar{x} = \Sigma\, l\, x / \Sigma\, l$, $\bar{y} = \Sigma\, l\, y / \Sigma\, l$; r_{max} = farthest weld end from G ; τ'' acts $\perp$ r.

Example 5 : (Fig 6) P = 10 kN, e = 150 mm, 2 welds l = 100, b = 80 : $A = 0.707\, s \times 200$ $\rightarrow \tau' = 70.7/s$. $J_u = 2\,[100^3/12 + 100 \times 40^2] = 4.87 \times 10^5$ mm³ $\rightarrow J = 344 \times 10^3\, s$. $T = 1.5 \times 10^6$ N·mm ; $r = \sqrt{(50^2 + 40^2)} = 64$ mm. $\tau'' = 1.5 \times 10^6 \times 64 / (344 \times 10^3\, s) = 279/s$. Components at the worst end : horizontal $279\,(50/64) = 218/s$, vertical $70.7 + 279\,(40/64) = 245/s$ $\rightarrow \tau = 328/s$.

τ_{allow} = 80 MPa → s = 328/80 = 4.1 mm → s = 5 mm (standard fillet size) ; minimum size also governed by plate thickness (≥ 3 mm for $t \le 6$).

Example 7 (bending) : bracket plate welded to a column by two horizontal fillet welds b = 100 mm long, placed d = 150 mm apart ; P = 8 kN at e = 200 mm. $M = 1.6\times10^6$ N·mm ; $Z_u = b\ d = 15000$ mm². $\sigma_b = M/(0.707\ s\ Z_u) = 151/s$. Direct shear $\tau' = 8000/(0.707\ s \times 200) = 56.6/s$. Resultant = $\sqrt{(151^2 + 56.6^2)}/s = 161/s$. With τ_{allow} = 80 → s = 2 mm ; but minimum practical size (thicker plate 10 mm → 5 mm) → s = 5 mm.

(c) Torsion of circular / welded tubular joints

Circular hollow shaft welded all round to a plate and twisted : $\tau = T\ d / 2J = 2T / (\pi\ d^2\ t_w)$ with $t_w = 0.707\ s$; combined with bending M : $\tau = \sqrt{(\tau_T^2 + \tau_b^2)}$, $\tau_b = 4M/(\pi\ d^2\ t_w)$.

Welding defects & comparison

- Defects : porosity (gas), slag inclusion, lack of fusion / penetration, undercut, cracks (hot / cold), overlap, distortion, residual stress → reduce static & fatigue strength ; detected by DPT, MPI, radiography, ultrasonic.

Point	Bolted	Welded
Permanence	detachable	permanent
Weight / cost	heavier (holes, plates, nuts)	lighter, cheaper
Fatigue	good with preload	poor at weld toe
Stress concentration	holes + threads	weld toe, residual stress
Skill / inspection	simple	skilled labour + NDT

Remember: Welded joints under fatigue are limited by weld toe : use stress range ($\Delta\sigma$) and K_f ; peen, grind toes, use full penetration welds to improve fatigue life. Weld size should not exceed the thinner plate thickness ; leg size ≤ 1.2 t, minimum 3-5 mm.

Topic	Key point
Preload	$F_i = 2840\ d$; $T = 0.2\ F_i\ d$
Joint constant	$C = k_b/(k_b+k_m)$; $F_b = F_i + C\ P$
Separation	$P_o = F_i/(1 - C)$
Eccentric (shear)	$F' = P/n$, $F'' = P\ e\ r/\Sigma r^2$, vector sum
Fillet weld	$P = 0.707\ s\ l\ \tau$ (×2 for double)
Eccentric weld	$\tau' = P/A$, $\tau'' = P\ e\ r/J$, $J = 0.707\ s\ J_u$

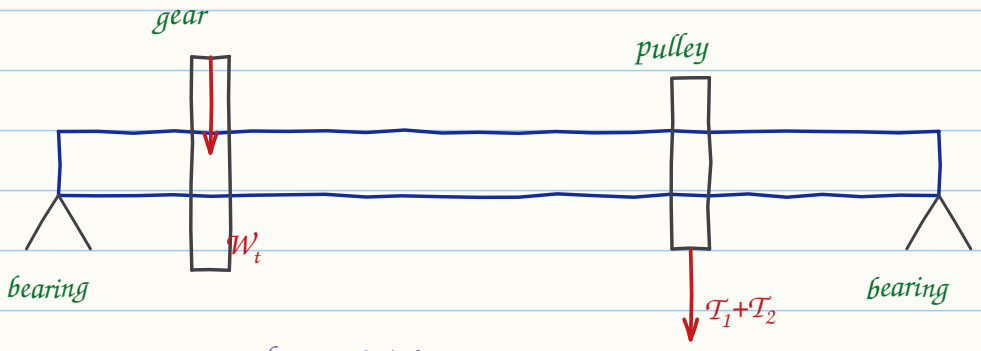
UNIT - III

Power Transmission Shafts and Couplings

1. Power Transmission Shafts

Shaft = rotating member (usually circular) that transmits power / torque and supports gears, pulleys, sprockets, flywheels. It is loaded by torsion, bending and sometimes axial load (helical gears, thrust).

- *Transmission shaft / line shaft* - carries power from the source to the machine ; *counter shaft* - intermediate shaft driven by a line shaft ; *spindle* - short shaft of a machine tool ; *axle* - supports rotating wheels, carries only bending (no torque).
- *Materials* : mild steel (C30, C40, 40C8, 45C8) - common ; alloy steel 40Cr1, 40Ni14, 35Mn2Mo28, 15Ni2Cr1Mo15 (case hardened) for high loads ; cold drawn / ground - better surface & accuracy up to 50 mm ; hot rolled - large sizes.
- *Requirements* : adequate strength (static & fatigue), rigidity (torsional & lateral), sizes from standard series : 6 8 10 12 14 16 18 20 22 25 28 30 32 36 40 45 50 55 60 65 70 75 80 90 100 mm...



$T = 9.55\times10^6\ P / N$ (N·mm)

M = max at gear / pulley section → shaft carries M and T together

Fig 1: A shaft with gear and pulley - bending + torque

Loads on the shaft

- *Torque* : $T = 60\ P / 2\pi N = 9.55\times10^6\ P(kW) / N(rpm)$ N·mm.
- *Gear* : tangential force $W_t = 2T/D$; radial $W_r = W_t \tan\varphi$ (spur) ; (helical : axial $W_a = W_t \tan\beta$ too).
- *Belt pulley* : $(T_1 - T_2)\ R = T$, bearing load = $T_1 + T_2$; $T_1/T_2 = e^{\mu\theta}$; for design $T_1 + T_2 \approx 1.5\text{-}2\ (T_1 - T_2)$ ... use given data.

- Resolve loads in vertical and horizontal planes, find reactions, draw B.M.D. in both planes, $M = \sqrt{(M_V^2 + M_H^2)}$ at the critical section ; draw T.M.D.

2. Design of Shafts - Strength Basis

(a) Shaft under torsion only

$$T = (\pi/16)\,\tau\, d^3 \quad \rightarrow \quad d^3 = 16\,T / \pi\,\tau_{allow}$$

hollow : $T = (\pi/16)\,\tau\, d_o^3\,(1 - K^4)$, $K = d_i/d_o$

- Allowable shear stress (ASME code) : $\tau_{allow} = 0.30\,S_y$ or $0.18\,S_{ut}$ (take the smaller). If a keyway is present reduce by 25% → $\tau_{allow} \times 0.75$.

Typical : 40C8 ($S_y = 330$, $S_{ut} = 580$) → $0.3 \times 330 = 99$; $0.18 \times 580 = 104$ → 99 MPa ; with keyway 74 MPa. For general shafting use τ = 40-60 MPa (machine-shop practice), small shaft with sudden shock : 20-30 MPa.

(b) Shaft under bending only (axle)

$$d^3 = 32\,M / \pi\,\sigma_{b,allow} \quad , \quad \sigma_{b,allow} = S_y/n \approx 0.6\,S_y / \ldots \text{(use given)}$$

(c) Combined bending and torsion - ASME code equation

Use maximum shear theory (ductile) with shock and fatigue factors K_b (bending) and K_t (torsion) which cover suddenly applied / reversing loads :

$$d^3 = 16 / (\pi\,\tau_{allow}) \cdot \sqrt{(\,(K_b\,M)^2 + (K_t\,T)^2\,)} \quad \text{(solid)}$$

$$d_o^3\,(1 - K^4) = 16 / (\pi\,\tau_{allow}) \cdot \sqrt{(\,(K_b\,M)^2 + (K_t\,T)^2\,)} \quad \text{(hollow)}$$

Type of load	K_b	K_t
Rotating shaft : gradually applied / steady	1.5	1.0
suddenly applied, minor shock	1.5 - 2.0	1.5 - 2.0
suddenly applied, heavy shock	2.0 - 3.0	1.5 - 3.0
Stationary shaft : gradually applied / steady	1.0	1.0
suddenly applied	1.5 - 2.0	1.5 - 2.0

Why : Rotating shaft → bending stress is completely reversed each revolution (fatigue !) so $K_b > 1$ even for steady load ; torque is steady → $K_t = 1$. That is the reason for $K_b = 1.5$, $K_t = 1.0$.

(d) With axial load F (tension / compression)

$$d^3 = 16 / (\pi \tau_{allow} (1 - K^4)) \cdot \sqrt{([K_b M + \alpha F d (1 + k^2)/8]^2 + (K_t T)^2)}$$

α = column factor = 1 for tension ; for compression of slender shaft $\alpha = 1/(1 - 0.0044 L/K)$ $(L/K < 115)$, K = radius of gyration. Axial load in short shafts is small - usually ignored. Equation is solved by trial (d appears on both sides) or neglect F first.

Example 1 : shaft transmits 30 kW at 360 rpm. Maximum bending moment 600 N·m (at gear). Material 40C8 (S_y = 330, S_{ut} = 580 MPa), keyway present. Rotating shaft, gradual load.

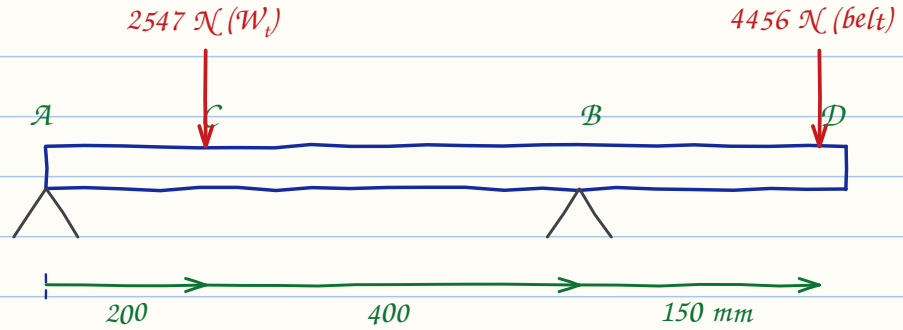
$T = 9.55\times10^6 \times 30/360 = 796\times10^3$ N·mm. τ_{allow} = min (0.3×330 , 0.18×580) = 99 → × 0.75 (keyway) = 74.3 MPa. K_b = 1.5, K_t = 1.

$d^3 = 16/(\pi \times 74.3) \sqrt{((1.5 \times 600\times10^3)^2 + (796\times10^3)^2)} = 0.0685 \times 1.2015\times10^6 = 82.4\times10^3$ → d = 43.5 mm → d = 45 mm (standard).

Hollow shaft (k = 0.5) : $d_o = d / (1 - K^4)^{1/3} = 43.5/0.979 = 44.4$ → provide d_o = 45 , d_i = 22.5 : weight ratio = $d_o^2 (1 - k^2) / d^2 = 0.78$ → 22% lighter for same strength. Hollow shaft has larger outer dia, hence higher cost / bearing size.

Worked example with gear and overhung pulley

Example 5 : 12 kW at 300 rpm. Gear (D = 300 mm, 20° pressure angle) at C, pulley (D = 400 mm, $T_1 = 2.5\ T_2$) overhung at D. Bearings A, B 600 mm apart. 40C8 with keyway (τ_{allow} = 74.3 MPa).



$T = 9.55\times10^6 \times 12/300 = 382\times10^3$ N·mm. Gear : $W_t = 2T/D = 2\times382\times10^3/300 = 2547$ N , $W_r = 2547 \tan 20° = 927$ N. Pulley : $T_1 - T_2 = 2T/D = 1910$ N → T_2 = 1273 , T_1 = 3183 → belt load $T_1 + T_2$ = 4456 N.

Take moments (V-plane) about A : $R_B \times 600 = 2547 \times 200 + 4456 \times 750$ → R_B = 6419 N , R_A = 584 N. B.M. at C = 584 × 200 = 117×10³ ; at B = 4456 × 150 = 668×10³ N·mm (overhang). H-plane : $M_C = 618\times200 = 124\times10^3$, M_B = 0. Resultant M_C =

170×10^3 N·mm , $M_B = 668\times10^3$ N·mm → section B is critical (torque also acts there).

Steady load ($K_b = 1.5$, $K_t = 1$) : $d^3 = 0.0685\sqrt{((1.5 \times 668\times10^3)^2 + (382\times10^3)^2)} = 73.5\times10^3$ → $d = 41.9$ mm → $d = 45$ mm.

Heavy shock ($K_b = 2.0$, $K_t = 1.5$) : $d^3 = 0.0685\sqrt{((2 \times 668\times10^3)^2 + (1.5 \times 382\times10^3)^2)} = 99.7\times10^3$ → $d = 46.4$ → $d = 50$ mm. Shock factors raised the size by 11%.

Example 5a (with axial load) : $M = 300$ N·m, $T = 200$ N·m, axial tension $F = 8$ kN, $\tau_{allow} = 60$ MPa, $K_b = 1.5$, $K_t = 1$, $\alpha = 1$, solid. Try $d = 35$: $F\,d/8 = 35\times10^3$ N·mm ; $K_b M = 450\times10^3$; $d^3 = (16/\pi 60)\sqrt{((450 + 35)^2 + 200^2)\times10^3} = 0.0849 \times 524.6\times10^3 = 44.5\times10^3$ → $d = 35.4$ mm → $d = 36$ mm (axial term raised d by < 2% - usually neglected for short shafts).

3. Shaft Design Based on Rigidity & Critical Speed

Torsional rigidity

$$\theta = T L / G J \text{ (rad)} \rightarrow \theta^\circ = 584\, T L / G d^4 \text{ (solid shaft)}$$
$$d^4 = 584\, T L / G \theta^\circ$$

- Allowable twist : 0.25°/m for machine tool shafts ; 0.5 - 1°/m for line shafts ; ≤ 1° per 20 d length for general transmission (precision drive).

Example 2 : same shaft ($T = 796$ N·m) , $L = 1$ m , $G = 80$ GPa , $\theta_{allow} = 0.5°/m$: $d^4 = 584 \times 796\times10^3 \times 1000 / (80\times10^3 \times 0.5) = 11.6\times10^6$ → $d = 58.4$ mm > 45 mm. So rigidity governs here → provide $d = 60$ mm. Always check both strength and rigidity.

Lateral deflection & critical speed

- Slope at bearings (≈ 0.001 rad for plain bearings ; 0.0005 for roller bearings) and deflection at gears (≤ 0.0005 L ..) maintain proper gear mesh and bearing life.
- Critical speed (whirling) : $\omega_c = \sqrt{(k/m)} = \sqrt{(g/\delta)}$, $N_c = 946/\sqrt{\delta}$ rpm (δ = static deflection in mm under the loads). Operating speed should be < 0.8 N_c or > 1.3 N_c ; rigid shaft (below) vs flexible shaft (above N_c).

Fatigue design (fluctuating loads)

Rotating shaft : bending completely reversed ($\sigma_a = 32\, M K_f/\pi\, d^3$, $\sigma_m = 0$) + steady torque ($\tau_m = 16\, T/\pi\, d^3$). Use Unit I expression (Soderberg / von Mises) :

$$d^3 = (32\ n / \pi)\sqrt{\left((K_f M / S_e)^2 + (3/4)(T / S_y)^2 \right)}$$

Example 5b (fatigue) : rotating shaft, $M = 500$ N·m (reversed), $T = 400$ N·m (steady), $S_y = 380$, corrected $S_e = 200$ MPa, $K_f = 1.8$, $n = 2$. $d^3 = (32\times2/\pi)\sqrt{\left((1.8\times500\times10^3/200)^2 + 0.75\,(400\times10^3/380)^2 \right)} = 20.37 \times 4591 = 93.5\times10^3 \rightarrow d = 45.4$ mm $\rightarrow$ **$d = 50$ mm** (standard, next size up).

- Steps : (1) T from P and N ; (2) forces from gears / pulleys ; (3) V and H planes, B.M.D. ; (4) critical section ; (5) choose material, τ_{allow} (or S_e, K_f) ; (6) d from ASME or Soderberg ; (7) check torsional / lateral rigidity ; (8) standard size, keyway, shoulders with fillet $r \geq 0.1\ d$; (9) check critical speed.
- Design details : stepped shoulders 5-10% larger ; relief grooves ; key seat sled-runner ; interference fits ; use $K_f = 1.6$-2.0 for keyway in bending, 1.3-1.5 for shoulder fillet.

4. Keys, Keyways and Splines

A **key** is a machine element placed in a keyway between shaft and hub to prevent relative rotation and transmit torque. It is the weakest link - by design (cheap, sheared in overload).

Key type	Features
Sunk - rectangular / square	$w = d/4$, $t = 2w/3$ (rect) , $t = w$ (square) ; most common
Gib-head taper (1 in 100)	hub driven onto key, can be removed ; gives axial hold
Woodruff	half-disc, sits deep in shaft, self-aligning ; machine tools, tapers
Feather / sliding	fixed to shaft, hub slides axially ; gear shifting
Splines (involute / straight)	many integral keys, high torque, hub slides ; gear box

$$\text{Shear : } T = l\ w\ \tau_k\ (d/2) \rightarrow \tau_k = 2T / (d\ l\ w)$$

$$\text{Crushing : } T = l\ (t/2)\ \sigma_c\ (d/2) \rightarrow \sigma_c = 4T / (d\ l\ t)$$

$$\text{Equal strength with shaft (square key) : } l = (\pi/2)\ d\ \tau_s/\tau_k \approx 1.57\ d$$

- Key stock : 40C8 / 45C8 ($S_y \approx 300$; $\tau_k = 0.5\ S_y/n$). Key is weaker than the shaft in crushing $\rightarrow$ l is kept $\approx$ hub length (1.25 d - 1.5 d). For longer hubs use 2 keys at 120°-180° (each counted as 0.75).

- Keyway reduces shaft torsional strength by ~ 25% (ASME) and gives $K_f \approx 2.0$ (bending) → use sled-runner ends and radiused corners.

Example 6 : $d = 45$ mm, $T = 796$ N·m, key $14 \times 9 \times 60$: $\tau_k = 2\times796\times10^3/(45\times60\times14) = 42$ MPa (OK) ($\leq 0.5\, S_y/n$ for 40C8 key $= 165/2 = 82$) ; $\sigma_c = 4\times796\times10^3/(45\times60\times9) = 131$ MPa $\leq S_y/n = 165$ (OK).

5. Couplings

A coupling connects two shafts end to end to transmit torque while allowing for assembly, alignment error, shock absorption, and disconnection (clutch can be engaged during running, coupling only at rest).

Class	Types	Feature
Rigid	sleeve (muff), split muff, clamp (compression), flange (protected / unprotected)	shafts must be collinear
Flexible	bushed-pin, Oldham, chain, gear, diaphragm, rubber-tyre, grid	allows small misalignment, absorbs shock
Universal	Hooke's (cardan) joint, double Hooke's	intersecting shafts, angle ≤ 30°

- Requirements : easy assembly / disassembly, positive torque transfer (no slip), minimal weight (inertia), safe (no projecting bolt heads - use protective rim), able to absorb vibration, low maintenance.

Design torque : $T_d = K_s\, T$ where service factor K_s = 1.0-1.5 (electric motor drive) to 2.0-2.5 (engine / heavy shock). Mostly the rated T is multiplied by 1.5 for design.

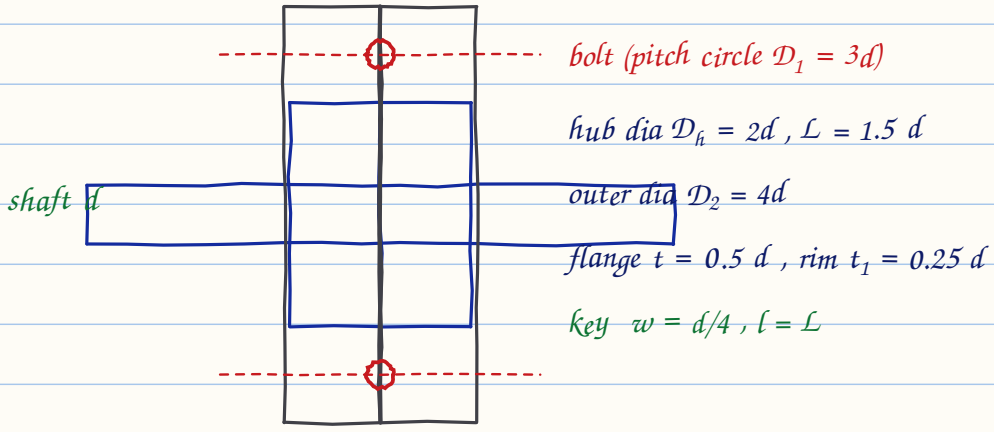
Comparison / selection of couplings

Coupling	Misalignment	Torque	Typical use
Muff / sleeve	none	low	light line shafts
Clamp (split muff)	none	low-medium	line shafting, easy removal
Flange	none (rigid)	high	turbines, heavy rigid drives
Bushed-pin	small angular & parallel	medium	motor - pump / fan
Oldham	parallel offset	low-medium	instrument, feed drives
Hooke's	angular ≤ 30°	medium-high	automobile propeller shaft
Gear / grid	1-3° angular	very high	rolling mills, heavy drives

- Catalogue selection : $T_{design} = K_s\, T_{rated}$; pick the smallest standard size with torque capacity ≥ T_{design} and bore range including d ; check max speed, misalignment and bore of both shafts.

6. Design of Flange Coupling

Two cast-iron (FG 200) flanges keyed to the shaft ends and held by n fitted bolts on a pitch circle. Standard proportions (IS 2949 / Bhandari) in terms of shaft dia d :



n = 3 (d ≤ 40) , 4 (40-100) , 6 (100-180)

Fig 2: Protected type flange coupling - standard proportions

Part	Dimension	Part	Dimension
Hub dia D_h	2 d	Bolt pitch circle D_1	3 d
Hub length L	1.5 d	Outer flange dia D_2	4 d
Flange thickness t	0.5 d	Protecting rim t_1	0.25 d
Key width w / thickness	d/4 ; 2w/3 (standard)	No. of bolts n	3, 4, 6

Design steps - check each part in shear (Torque T basis)

1. Shaft : $d^3 = 16\ T / \pi\ \tau_s$
2. Hub (hollow shaft in torsion) : $T = (\pi/16) \cdot (D_h^4 - d^4)/D_h \cdot \tau_h$
3. Key : shear $T = l\ w\ \tau_k\ d/2$, crushing $T = l\ (t/2)\ \sigma_c\ d/2$
4. Bolts (shear) : $T = n\ (\pi/4)\ d_b^2\ \tau_b \cdot D_1/2$
5. Flange at hub junction (shear) : $T = \pi\ D_h\ t\ \tau_f \cdot D_h/2$
6. Bolt crushing : $T = n\ d_b\ t\ \sigma_c \cdot D_1/2$

- Hub : with $D_h = 2d$: $T = 1.47\ d^3\ \tau_h$; flange : $T = 2\pi\ d^2\ t\ \tau_f$. Bolts are turned & fitted in reamed holes so that they are in shear, not tension (clearance bolts rely on friction - poor).

- Material : shaft & key steel (45C8 , 30C8) ; flange C.I. FG 200 ($\tau \approx 15$ MPa) ; bolts mild steel Fe 410. Typical allowable : τ_s = 40-60 , τ_{bolt} = 30 , τ_{key} = 40-50 , σ_c (key) = 80-100 , $\tau_{C.I.}$ = 15 MPa.

Example 3 : Flange coupling to transmit 15 kW at 600 rpm. $T = 9.55\times10^6 \times 15/600 = 238.75\times10^3$ N·mm. τ_s = 40 MPa.

Shaft : $d^3 = 16 \times 238750 / (\pi \times 40) = 30.4\times10^3 \rightarrow d = 31.2 \rightarrow$ 32 mm. Then $D_h = 64$, $L = 48$, $D_1 = 96$, $D_2 = 128$, $t = 16$, $t_1 = 8$ mm , $n = 3$.

Bolts : $238750 = 3 \times (\pi/4)\, d_b^2 \times 30 \times 48 \rightarrow d_b^2 = 70.4 \rightarrow d_b = 8.4$ mm → M10 fitted bolts. Key 10 × 8 × 48 : $\tau = 2T/(d\, l\, w) = 2\times238750/(32\times48\times10) = 31$ MPa < 40 (OK). Hub $\tau_h = 238750/(1.47 \times 32^3) = 5$ MPa (OK) ; flange $\tau_f = 238750/(2\pi \times 32^2 \times 16) = 2.3$ MPa (OK).

7. Muff (Sleeve) and Clamp Couplings - Rigid Types

Sleeve / muff coupling

A hollow cast-iron cylinder (muff) fits over the two shaft ends and is keyed to both ; simplest rigid coupling for light loads. Proportions : outer dia $D = 2d + 13$ mm ; length $L = 3.5\, d$; key along the full half-length ($L/2$ each shaft).

Muff (hollow shaft in torsion) : $T = (\pi/16) \cdot (D^4 - d^4)/D \cdot \tau_c$

Key (each shaft) : $\tau_k = 2T / (d\, (L/2)\, w) \le \tau_{allow}$

Split-muff / clamp (compression) coupling

Muff split into two halves clamped on the shafts by n bolts ; torque is transmitted by friction (a key is only for safety). No need to move shafts axially → easy mounting. Clamp bolts of total tension n W press the halves on the shaft :

$p = n\, W / (d\, L)$ (contact pressure on projected area)

$T = \mu\, p\, (\pi\, d\, L)\, (d/2) = (\pi/2)\, \mu\, n\, W\, d$

- $\mu \approx 0.25$ (steel on C.I.) ; bolt tension $W = \sigma_t\, A_t$ (core area) ; proportions : $D = 2d + 13$, $L = 3.5\, d$, n = 4-8 bolts. Use key as well if reversing torque or shock.

Example 7 : d = 40 mm, T = 400 N·m, μ = 0.25, 6 bolts : $W = 2T / (\pi\, \mu\, n\, d) = 2\times400\times10^3/(\pi \times 0.25 \times 6 \times 40) = 4244$ N per bolt → $A_t = 4244/60 = 71$ mm² (σ = 60 MPa) → M12 bolts.

8. Bushed-Pin Flexible Coupling

Modified flange coupling : the bolts are replaced by pins fitted with rubber bushes (leather / neoprene) on one flange ; the bush absorbs shock & allows ≈ 0.5-3° misalignment and axial float. Used for motor-pump, fan, compressor drives.

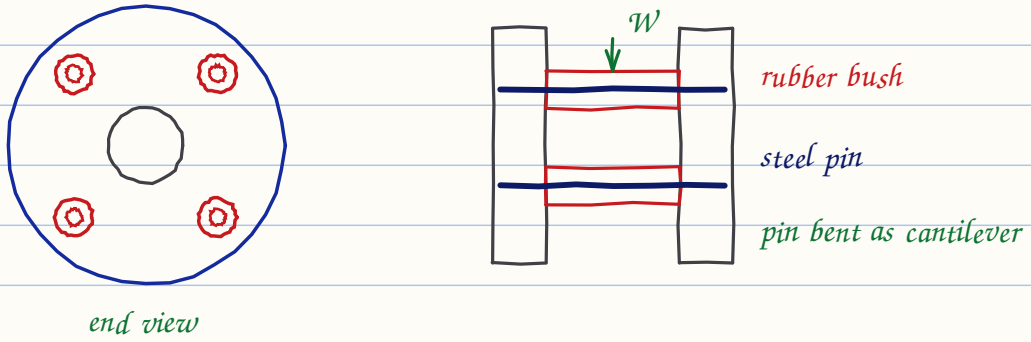


Fig 3: Bushed-pin flexible coupling

- Bush bearing pressure : $p_b = W / (d_2\, l) \le 0.5\ N/mm^2$ (rubber) ; W = load per pin = $2T / (n\, D_1) = 2T/(n\, D_p)$ where D_p = pitch circle dia of pins.
- Pin is treated as a cantilever loaded at the middle of the bush (length l) with a clearance $c \approx 5$ mm between flanges :

$$M = W\,(l/2 + c)$$

$$\sigma_b = 32\, M / \pi\, d_1^3 \le 25\text{-}35\ MPa$$

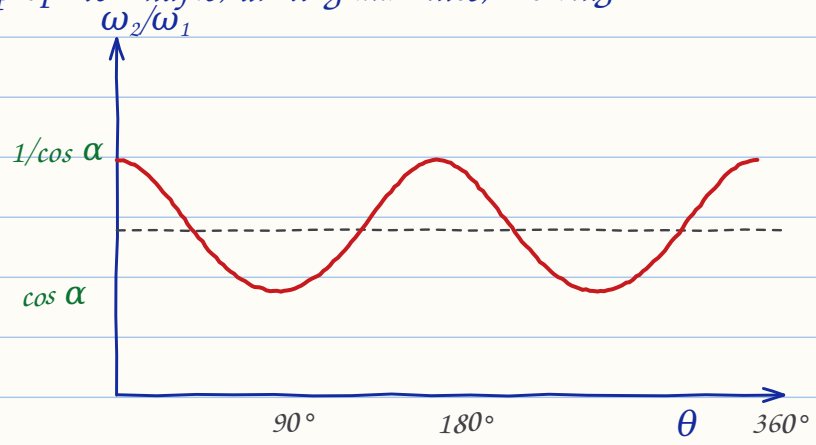
$$\tau = 4\, W / \pi\, d_1^2 \quad ; \quad \sigma_{eq} = \sqrt{(\sigma_b^2 + 3\tau^2)}$$

- Torque capacity : $T = W\, n\, D_p/2 = p_b\, d_2\, l\, n\, D_p/2$. Hub, key, shaft designed as in flange coupling. Number of pins 4-8 ; $d_2 = d_1 + 2$ (sleeve + rubber ≈ 6) ; pin is tapered on the driving flange side, nut & washer on the other.

Example 4 : Same $T = 238.75\times10^3$ N·mm, $d = 32$ mm. Take $n = 4$ pins on $D_p = 150$ mm : $W = 2T/(n\, D_p) = 2\times238750/(4\times150) = 796$ N. Bush $d_2 = 32$, $l = 50 \rightarrow p_b = 796/(32 \times 50) = 0.50\ N/mm^2$ (OK). $M = 796 \times (25 + 5) = 23.9\times10^3$ N·mm ; $\sigma_b = 30 \rightarrow d_1^3 = 32 \times 23900 / (\pi \times 30) = 8.1\times10^3 \rightarrow d_1 = 20$ mm ; (bush : $20 + 2\times6 = 32$ (OK)). $\tau = 4\times796/(\pi\, 20^2) = 2.5$ MPa - negligible.

9. Universal (Hooke's) Coupling

Joins two shafts whose axes intersect at an angle α (usually ≤ 20°, max 30-35°) : two forked yokes joined through a cross (spider) with four trunnions ; used in automobile propeller shafts, drilling machines, steering.



α = 30° : 0.866 ≤ ω_2/ω_1 ≤ 1.155

$\omega_2/\omega_1 = \cos\alpha / (1 - \sin^2\alpha \cos^2\theta)$

two cycles per revolution

Fig 4: Velocity ratio of a single Hooke's joint

- Even if shaft 1 runs at constant ω_1, driven shaft 2 has varying speed : $\omega_2 = \omega_1 \cos\alpha / (1 - \sin^2\alpha \cos^2\theta)$, θ = angle turned by driver from the position when the fork pins lie in the plane of shafts.
- Max $\omega_2 = \omega_1 / \cos\alpha$ (θ = 0°, 180°) ; min $\omega_2 = \omega_1 \cos\alpha$ (θ = 90°, 270°) ; speed fluctuation = $\omega_1 (1/\cos\alpha - \cos\alpha) = \omega_1 \sin\alpha \tan\alpha$. Equal at θ where $\tan\theta = \pm 1/\sqrt{(\cos\alpha)}$.
- Torque transmitted varies inversely : $T_2 = T_1 \omega_1/\omega_2$. Fluctuating speed → angular acceleration, torque ripple, vibration - gets worse as α increases.
- Double Hooke's joint (intermediate shaft) : angles at both joints equal and fork planes of the two joints in the same plane → the speed variation of the first joint is cancelled → $\omega_3 = \omega_1$ (constant velocity drive).

Design of the joint

Shaft : $d^3 = 16\,T / \pi\,\tau_s$

Pin (cross arm) load : $W = T / 2\,a$ (a = distance of trunnion from axis)

Pin : bearing $p = W / d_1\,l_1$; bending $M = W\,l_1/2 \rightarrow \sigma_b = 32\,M / \pi\,d_1^3$; shear $\tau = 4W$

- Yoke ears, boss & cross are checked in shear and tension ; use $T_{max} = T_1/\cos\alpha$ at the worst position ; pins are of 40C8 / case hardened, bearings needle roller or bushes.

Example 8 : Hooke's joint with $\alpha = 20°$, driving shaft at 300 rpm : $N_{2,max} = 300/\cos 20° = 319.2$ rpm ; $N_{2,min} = 300 \cos 20° = 281.9$ rpm ; variation = 37.3 rpm (≈ 12%). With a double joint (equal angles) → 300 rpm constant.

Example 9 : 10 kW at 300 rpm, $\tau_s = 40$ MPa : $T = 318\times10^3$ N·mm → $d^3 = 16\,T/\pi\,\tau = 40.5\times10^3$ → $d = 34.4$ → 35 mm. Trunnion arm at $a = 50$ mm → $W = T/2a = 3183$ N. Pin $d_1 = 18$ mm, $l_1 = 24$ mm : $p = W/d_1 l_1 = 7.4$ MPa (OK) ; $M = W\,l_1/2 = 38.2\times10^3$ → $\sigma_b = 32\,M/\pi\,d_1^3 = 67$ MPa (OK) ; $\tau = 4W/\pi\,d_1^2 = 12.5$ MPa (OK).

Other flexible couplings (short note)

- Oldham : two slotted discs + central disc with perpendicular tongues - parallel (offset) shafts ; speed constant, wear in slots.
- Gear (toothed) coupling : external teeth hubs in internal-tooth sleeve - high torque, allows 1-3° misalignment (steel mills).
- Chain : two sprockets enclosed by a double-strand chain - simple, cheap. Diaphragm / disc : metallic flexing discs - no lubrication, zero backlash.

Typical exam questions - Unit III

- (1) Derive the ASME code equation for a shaft under bending and torsion with shock factors ; state K_b, K_t values.
- (2) Design a shaft to transmit P kW at N rpm carrying a gear and pulley - draw B.M.D. in both planes.
- (3) Solid vs hollow shaft : compare strength, weight and torsional rigidity.
- (4) Design a protected-type flange coupling for given P, N ; sketch with dimensions.
- (5) Design a bushed-pin coupling ; why is rubber bush used ? What is checked in the pin ?
- (6) Hooke's joint : derive velocity ratio, explain why a double joint gives constant speed.
- (7) Differences between rigid and flexible coupling ; between clutch and coupling.

Item	Key formula
Torque	$T = 9.55\times10^6\ P / N$ (N·mm)
Shaft (ASME)	$d^3 = 16\sqrt{((K_bM)^2 + (K_tT)^2)}/(\pi\ \tau)$
Twist	$\theta° = 584\ T\ L / G\ d^4$
Flange bolts	$T = n\ (\pi/4)\ d_b^2\ \tau_b\ D_1/2$
Bush pressure	$p_b = W/(d_2\ l) \le 0.5$ MPa
Hooke's joint	$\omega_2/\omega_1 = \cos\alpha/(1 - \sin^2\alpha\ \cos^2\theta)$

UNIT - IV

Design of Clutches, Brakes and Springs

1. Friction Clutches

A clutch connects / disconnects a driven shaft from a driving shaft (engine - gearbox) while running, by friction between surfaces pressed together by an axial force W. It must engage smoothly (slip, then lock), transmit the full torque without slip, and dissipate heat.

- Types : positive (jaw, dog, spline - no slip, engage at rest) ; friction (disc / plate - single & multi plate, cone, centrifugal) ; hydraulic / fluid (fluid coupling) ; electromagnetic.
- Friction material : lining woven asbestos-free / resin-bonded organic (μ = 0.3-0.4), sintered metal (0.15-0.4), cork (0.3-0.35, wet), leather (0.3-0.5) on C.I. or steel ; p_{max} = 0.1-0.5 N/mm² (dry), up to 1 N/mm² (wet).
- Requirements : high & stable μ, wear and heat resistance, good thermal conductivity, cheap, no fade, quick cooling, small inertia, balanced.

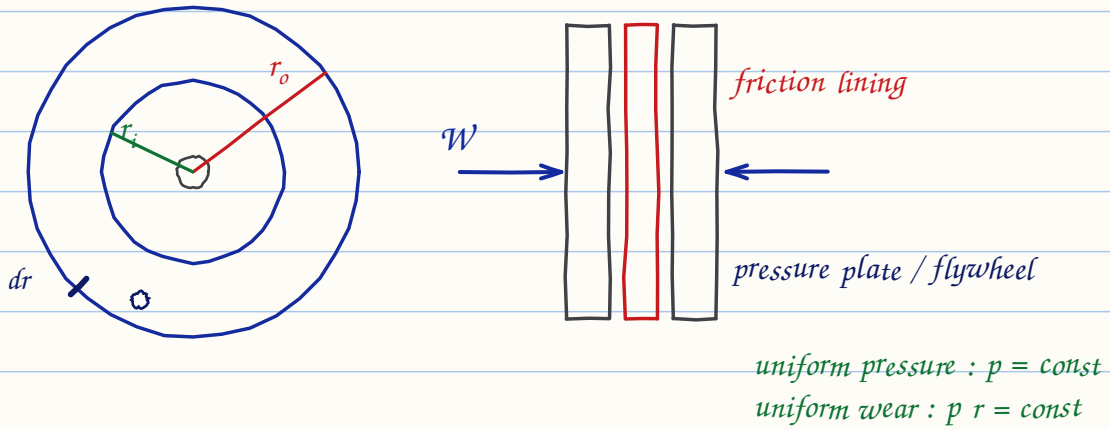


Fig 1: Single-plate (disc) clutch and pressure laws

Torque transmitting capacity - disc clutch

Annular lining : outer radius r_o, inner r_i, axial force W, coefficient μ. Take an element of radius r, width dr. Normal force $dW = p\ 2\pi\ r\ dr$; friction torque $dT = \mu\ p\ 2\pi\ r^2\ dr$.

(a) Uniform pressure theory (new clutch) : p = const

$$W = \pi\, p\, (r_o^2 - r_i^2)$$
$$T = (2/3)\,\pi\,\mu\, p\, (r_o^3 - r_i^3) = \mu\, W\, R_m$$
$$R_m = (2/3)\,(r_o^3 - r_i^3)/(r_o^2 - r_i^2)$$

(b) Uniform wear theory (worn-in clutch) : $p\,r = C$ = const

Wear $\propto p \times v \propto p\,r$ (since $v = \omega\, r$), so for equal wear $p\,r = C$; maximum pressure is at the inner radius : $C = p_{max}\, r_i$.

$$W = 2\pi\, C\, (r_o - r_i) = 2\pi\, p_{max}\, r_i\, (r_o - r_i)$$
$$T = \pi\,\mu\, C\, (r_o^2 - r_i^2) = \mu\, W\, R_m\,,\quad R_m = (r_o + r_i)/2$$

- For the same W : T(uniform pressure) > T(uniform wear) (R_m larger) , so a clutch is designed with the uniform wear value (lower, conservative) as the lining wears after the initial run-in.
- Multi-plate clutch : n_1 discs on driving shaft , n_2 on driven → number of pairs of contact surfaces $n = n_1 + n_2 - 1$: $T = n\,\mu\, W\, R_m$. Large torque in small diameter (motorcycles, wet clutch in oil, automatic transmission).
- For maximum torque (given r_o, p_{max}, uniform wear) : $dT/dr_i = 0 \rightarrow r_i = r_o/\sqrt{3} = 0.577\, r_o$. Usually $r_i/r_o = 0.6$-0.7 (practical).
- Power : $P = T\,\omega = 2\pi\, N\, T / 60$; design torque $T_d = K_s \times T_{engine}$ ($K_s = 1.25$-2.0). Engaging spring force = W (diaphragm spring in cars).

Example 1 : Single-plate clutch with both sides effective. $r_o = 125$ mm, $r_i = 75$ mm, $\mu = 0.3$, $p_{max} = 0.1$ N/mm², 1500 rpm. Uniform wear : $W = 2\pi \times 0.1 \times 75 \times 50 = 2356$ N ; $R_m = 100$ mm ; $T = n\,\mu\, W\, R_m = 2 \times 0.3 \times 2356 \times 100 = 141 \times 10^3$ N·mm = 141 N·m → $P = 141 \times (2\pi \times 1500/60) = 22.2$ kW.

(Uniform pressure, same p : $W = \pi \times 0.1 \times (125^2 - 75^2) = 3142$ N ; $R_m = 102$ mm ; $T = 2 \times 0.3 \times 3142 \times 102 = 192$ N·m - 36% more - not safe to use.)

Heat generated & slip time (clutch engagement)

$$\text{Slip energy}\quad E = \tfrac{1}{2}\, I_e\, \omega_1^2\,,\quad I_e = I_1\, I_2/(I_1 + I_2)$$
$$\text{Slip time}\quad t = I_2\, \omega_1 / (T - T_{load})$$
$$\text{Heat flux}\quad q = E / (A\, t)\quad (W/mm^2)$$

- Smooth engagement needs t = 0.5-1.5 s for cars. Too short → jerk (shock load on shaft), too long → overheating of lining (fade, glazing). Limit $q \approx 0.4\text{-}1.0\ W/mm^2$ for dry linings. Temperature rise $\Delta T = E / (m\ c)$.
- Positive (jaw / claw) clutch : square or spiral jaws ; torque by shear / bearing on the jaw face : $T = n\ (A_{jaw}\ \tau)\ R_m$; engaged only at rest or very low relative speed (no slip - shock).

2. Cone Clutch

Friction surfaces are conical (semi-cone angle $\alpha \approx 10°$ to $12.5°$; $\alpha > 8.5°$ so that the cone does not jam / self-lock : $\tan \alpha > \mu$). Wedge action gives large normal force with small axial W :

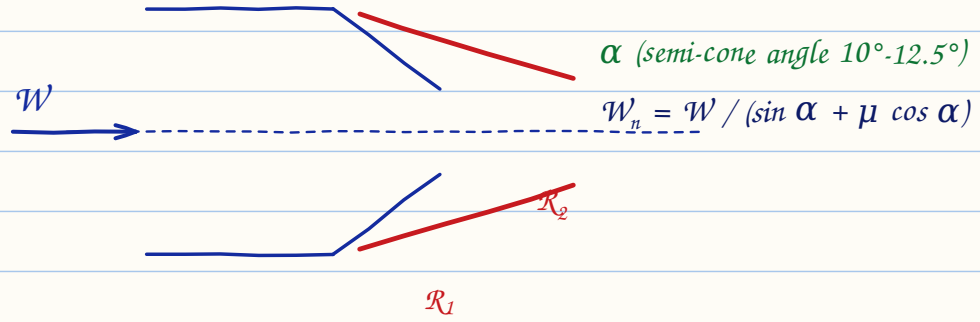


Fig 2: Cone clutch

Engagement : $W = W_n\ (\sin \alpha + \mu \cos \alpha)$ (W_n = normal force on cone)

Running (uniform wear) : $T = \mu\ W_n\ R_m = \mu\ W\ R_m / \sin \alpha$

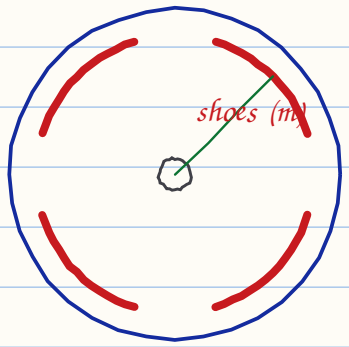
Uniform pressure : $R_m = (2/3)(R_1^3 - R_2^3)/(R_1^2 - R_2^2)$; $W = \pi\ p\ (R_1^2 - R_2^2)$

Uniform wear : $R_m = (R_1 + R_2)/2$

- Face width b along the cone ; $R_1 - R_2 = b \sin \alpha$. Pressure $p = W_n / (2\pi\ R_m\ b)$. Torque capacity ≈ 5-8 times more than a plate clutch of equal size for the same W ($\alpha = 12.5°$) - but needs accurate alignment, sticks if α too small, disengagement needs force W_{dis}.

3. Centrifugal Clutch

Shoes with friction lining are pivoted on a driving spider and are held inward by springs. When the speed rises above the engagement speed ω_1 the centrifugal force overcomes the spring and the shoes press on the drum - automatic clutch (mopeds, chain-saws, go-karts, starting of motors).



at speed ω :

centrifugal $P_c = m\,\omega^2\,r$

spring $P_s = m\,\omega_1^2\,r$ (ω_1 = engagement)

normal $N = P_c - P_s$, $F = \mu N$

$T = n\,\mu\,(P_c - P_s)\,R$

Fig 3: Centrifugal clutch

$P_c = m\,\omega^2\,r$, $P_s = m\,\omega_1^2\,r$ (r = c.g. radius of shoe when engaged)

Normal on drum : $N = P_c - P_s = m\,r\,(\omega^2 - \omega_1^2)$

$T = n\,\mu\,N\,R = n\,\mu\,m\,r\,(\omega^2 - \omega_1^2)\,R$

- Usually the shoes begin to engage at $\omega_1 = 0.75\,\omega$ (¾ running speed) → $P_s = (9/16)\,P_c$; $N = (7/16)\,m\,r\,\omega^2$. Pressure on lining : $p = N / (b\,l)$, $l = R\theta$ (shoe arc) , $p \le 0.1$-0.2 N/mm².

Example 2 : 15 kW, 900 rpm, 4 shoes, drum R = 150 mm, shoe c.g. r = 120 mm, μ = 0.25, $\omega_1 = 0.75\,\omega$. $T = 9550 \times 15/900 = 159$ N·m. $N = T/(n\,\mu\,R) = 159/(4 \times 0.25 \times 0.15) = 1061$ N. With $\omega = 94.25$ rad/s : $N = (7/16)\,m\,r\,\omega^2 \rightarrow m = 1061/(0.4375 \times 0.12 \times 8883) = 2.27$ kg per shoe. If $\theta = 60°$, b = 40 mm : l = 157 mm → $p = 1061/(157 \times 40) = 0.17$ N/mm² (OK).

4. Brakes - General

A brake absorbs the kinetic / potential energy of a moving system by friction and dissipates it as heat, to slow or stop (or hold) the system.

Energy to be absorbed : $E = \tfrac{1}{2}\,I\,(\omega_1^2 - \omega_2^2) + \tfrac{1}{2}\,m\,(v_1^2 - v_2^2) + m\,g\,h$

Braking torque : $T_b = \mu\,N\,r_d$; heat generated $Q = T_b\,\theta$ (θ = angle turned in braking)

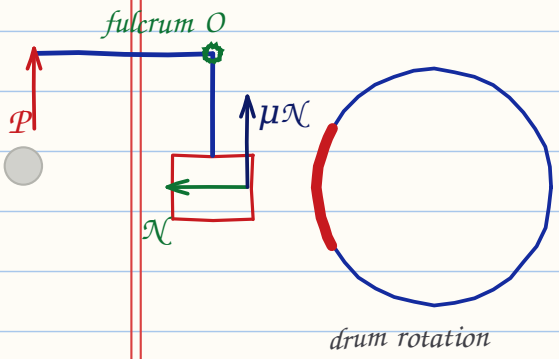
Temperature rise : $\Delta T = E / (m_{drum}\,c)$ ($c \approx$ 460-500 J/kg K steel / C.I.)

Brake	Description	Use
Block / shoe (external)	single / double rigid or pivoted shoe pressed on drum	railway, cranes, lifts
Band	flexible steel band with lining wrapped on drum	hoists, excavators, chain saw
Internal expanding	two shoes inside drum, expanded by cam / hydraulic cylinder	car, motorcycle rear
Disc (caliper)	pads squeezed on a rotating disc	cars, aircraft, bikes
Cone / Axial	like cone clutch	winches

- Brake materials : drum C.I. FG 260 ; shoe lining asbestos-free Ferodo / woven cotton / sintered, μ = 0.3-0.45, p_{max} = 0.3-1 N/mm² ; band : mild steel, lining leather / fabric. Brake must also be cooled (fins, ventilation).

5. Block (Shoe) Brake

Single block brake : block pressed on drum by lever ; normal reaction N, friction $F = \mu N$ tangential, braking torque $T_b = \mu N r$.



$P l = N a \mp \mu N c$

friction moment aids P → self-energizing

if $a \le \mu c$ → $P \le 0$: self-locking

(c = offset of fulcrum from the friction line)

Fig 4: Single block brake with offset fulcrum

$P l = N a - \mu N c$ (friction moment assists P) → self-energizing

$P l = N a + \mu N c$ (rotation reversed → friction opposes)

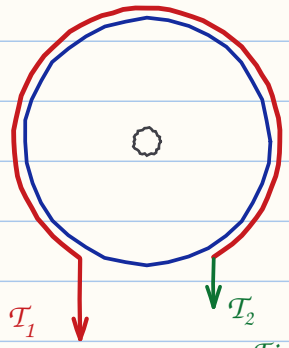
$T_b = \mu N r$, $N = P l / (a \mp \mu c)$

- Self-energizing brake : friction force itself helps to apply the brake → smaller P for the same torque (mechanical advantage ↑). Self-locking : when $a \le \mu c$, P becomes zero or negative - the brake grabs & cannot be released (used deliberately in some ratchets, but dangerous in vehicles).
- Pivoted block ($\theta > 60°$) : equivalent coefficient $\mu' = 4 \mu \sin(\theta/2) / (\theta + \sin\theta)$ (θ in rad) ; pressure uniform, no tilting moment. Double-block brake balances the shaft bearing load (no side thrust).

- Block pressure $p = N / (w \times 2\,r \sin(\theta/2)) \le 0.5\text{-}1\ N/mm^2$; heat per unit area per second $\mu\ p\ v \le 0.3\text{-}0.5\ N{\cdot}m/mm^2{\cdot}s$ (limit of heating).

6. Band Brake

The flexible band on drum : tension T_1 (tight) and T_2 (slack), with the Eytelwein relation for belt friction :



$T_1/T_2 = e^{\mu\theta}$

$T_b = (T_1 - T_2)\ r$

$p_{max} = 2\ T_1 / (w\ D)$

simple : $P\ l = T_2\ b$ (slack end on lever)

differential : $P\ l = T_2\ b_2 - T_1\ b_1$

Fig 5: Band brake - simple and differential

- Drum radius effective $r = r_d + t/2$ (t = band thickness). Wrap $\theta = 180°\text{-}270°$, $\mu = 0.2\text{-}0.4$ (leather / fabric on steel).
- Simple band : lever force $P = T_2\ b / l$ when slack end is on the lever (large wrap helps) ; if tight end T_1 on lever : $P = T_1\ b / l$ (twice or more).
- Differential band : tight and slack ends on opposite sides of fulcrum ; $P\ l = T_2\ b_2 - T_1\ b_1 = T_2\ (b_2 - b_1\ e^{\mu\theta}) \rightarrow$ self-locking if $b_2 \le b_1\ e^{\mu\theta}$; low P but unsafe for reversal.
- Band design : width w and thickness t from tension ; $\sigma_t = T_1 / (w\ t) \le 50\text{-}80$ MPa (with rivet-hole weakening : $T_1/((w - n\ d_r)\ t)$) ; max pressure at tight end $p_{max} = 2\ T_1/(w\ D) \le 0.3\text{-}0.5\ N/mm^2$.

Example 3 : Drum $D = 400$ mm, $T_b = 400\ N{\cdot}m$, wrap 240° (4.19 rad), $\mu = 0.25$. $e^{\mu\theta} = e^{1.047} = 2.85$. $T_1 - T_2 = 400/0.2 = 2000\ N \rightarrow T_2 = 2000/1.85 = 1082\ N$, $T_1 = 3082\ N$. Simple band, $b = 60$ mm, $l = 600$ mm : $P = T_2\ b / l = 1082 \times 60/600 =$ 108 N. Band $w = 50$, $t = 3$: $\sigma = 3082/(50 \times 3) = 20.5$ MPa (OK) ; $p_{max} = 2 \times 3082/(50 \times 400) = 0.31\ N/mm^2$ (OK).

7. Internal Expanding Shoe & Disc Brake

Internal expanding shoe brake (drum brake)

- Shoe pivoted at one end, expanded against rotating drum by cam / wheel cylinder. *Leading shoe* : drum drags shoe into the drum - friction moment M_f assists (self-energizing) ; *trailing shoe* : friction opposes.

$$\text{Actuating force : } F = (M_N - M_f)/c \quad \text{(leading)} \;;\; F = (M_N + M_f)/c \quad \text{(trailing)}$$
$$\text{Torque per shoe : } T = \mu\, p_{max}\, b\, r^2 (\cos\theta_1 - \cos\theta_2)/\sin\theta_a$$

- M_N = moment of normal forces about pivot, M_f = moment of friction forces ; self-locking when $M_f \geq M_N$. Pressure varies as $\sin\theta$ (zero at the pivot). Dual-leading arrangement (2 leading shoes) increases torque ; hydraulic cylinder gives equal force on both shoes.

Disc brake

Annular pads (sector of angle θ) clamped on a rotating disc by caliper pistons ; both sides of the disc work. Uniform wear used for design :

$$F = p_{max}\, r_i\, \theta\, (r_o - r_i) \quad \text{(per pad)}$$
$$T_b = n\,\mu\,F\,R_m = n\,\mu\,p_{max}\,\theta\,r_i\,(r_o^2 - r_i^2)/2\;,\quad R_m = (r_o + r_i)/2$$

- *Advantages* over drum : no fade (open to air, self-cleaning, ventilated discs), linear torque vs pedal force, uniform wear, easy pad change, stable in wet, balanced axial thrust. *Disadvantages* : needs high hydraulic pressure (servo), pad area small → pad wear faster, costly.

Example 4 : A 1000 kg vehicle at 72 km/h (20 m/s) stops. $E = ½ \times 1000 \times 20^2 = 200$ kJ. If the four brake drums have 20 kg mass each (80 kg total, $c = 480$ J/kg K) → $\Delta T = 200000/(80 \times 480) = 5.2$ °C for a single stop ; 25 repeated stops → fade. Disc pad ($n = 2$, $\mu = 0.35$, $p_{max} = 1$ N/mm², $\theta = 60° = 1.05$ rad, $r_o = 120$, $r_i = 70$) : $F = 1 \times 70 \times 1.05 \times 50 = 3675$ N ; $T = 2 \times 0.35 \times 3675 \times 95 = 244$ N·m per disc.

8. Helical Springs - Compression & Tension

A *spring* stores energy and releases it (shock absorption, force application, energy storage, measurement). Helical spring of round wire is loaded axially - the wire experiences *torsion + direct shear*.

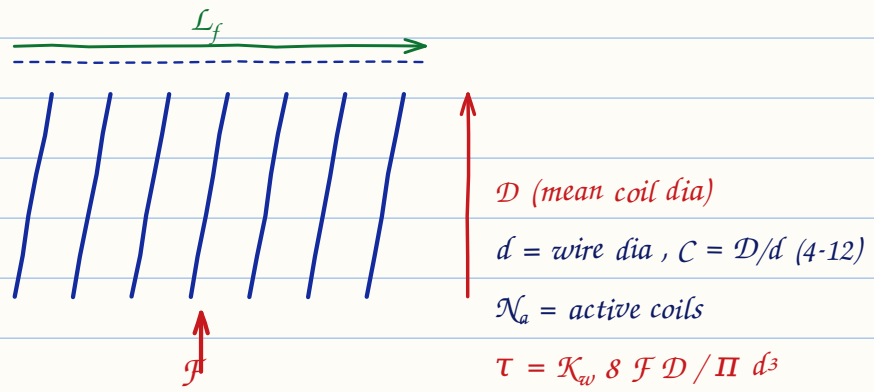


Fig 6: Helical compression spring terminology

- Spring index $C = D / d$ (4-12 preferred ; < 4 hard to coil, > 12 tangles / buckles). Active coils N_a ; total N_t ; pitch p ; free length L_f ; solid length L_s ; stiffness (rate) $k = F/\delta$.

End type	N_t	Solid length L_s
Plain ends	N_a	$(N_t + 1)\ d$
Plain & ground	$N_a + 1$	$N_t\ d$
Squared (closed)	$N_a + 2$	$(N_t + 1)\ d$
Squared & ground	$N_a + 2$	$N_t\ d$

$$\tau = K_w \cdot 8\ F\ D\ /\ \pi\ d^3 = K_w\ 8\ F\ C\ /\ \pi\ d^2$$
$$K_w = (4C - 1)/(4C - 4) + 0.615/C \quad \text{(Wahl factor)}$$
$$\delta = 8\ F\ D^3\ N_a\ /\ G\ d^4 = 8\ F\ C^3\ N_a\ /\ G\ d$$
$$k = F/\delta = G\ d^4\ /\ 8\ D^3\ N_a \quad , \quad U = \tfrac{1}{2}\ F\ \delta$$

- Wahl factor K_w accounts for direct shear $(1 + 0.5/C)$ and curvature of the wire (higher stress on the inside of the coil). K_w = 1.1-1.6 for C = 12-4. For static load on ductile wire $K_s = 1 + 0.5/C$ may be used ; for fatigue always K_w.
- Free length : $L_f = L_s + \delta_{max}$ + clearance (≈ 15% δ_{max} or 1-2 mm per coil). Buckling check : L_f/D < 2.6 (ends fixed, no guide) ; otherwise use a guide rod / tube. Surge : natural frequency $f_n = \tfrac{1}{2}\sqrt{(k / m_s)} \geq 13 \times$ operating frequency.
- Materials : hard-drawn (A227), oil-tempered (A229), music wire (A228 - best, $d \leq 6$ mm), chrome-vanadium (A231, high temp / fatigue), 55Si2Mn90 / 60Si2Mn (IS) for heavy springs, phosphor bronze, stainless. $\tau_{allow} = 0.45\text{-}0.5\ S_{ut}$ (static) ; G = 80-84 GPa (steel).

$$d = \sqrt{(8\,K_w\,F\,C/\pi\,\tau_{allow})} \quad ; \quad N_a = \delta\,G\,d/(8\,F\,C^3)$$ *(design steps)*

Example 5 : Spring for $F = 1000$ N and $\delta = 25$ mm, $C = 6$, $\tau_{allow} = 420$ MPa, $G = 84$ GPa. $K_w = 23/20 + 0.615/6 = 1.253$. $d = \sqrt{(8 \times 1.253 \times 1000 \times 6/(\pi \times 420))} = 6.75 \rightarrow d = 7$ mm. $D = 6 \times 7 = 42$ mm. $N_a = 25 \times 84000 \times 7/(8 \times 1000 \times 216) = 8.5 \rightarrow 9$; squared & ground $N_t = 11$, $L_s = 11 \times 7 = 77$ mm , $L_f = 77 + 25 + 4 = 106$ mm , $k = 40$ N/mm.

Springs in combination

Parallel (concentric) : $k = k_1 + k_2$, $F = F_1 + F_2$, same δ

Series : $1/k = 1/k_1 + 1/k_2$, same F , $\delta = \delta_1 + \delta_2$

- *Concentric springs* : same free length & deflection, opposite hand (no jamming), $D_{outer,inner}$ chosen with radial clearance ; $F_1/F_2 = k_1/k_2$; for equal τ, C equal $\rightarrow$ $d_1/d_2 = D_1/D_2$.

Fluctuating load (fatigue of springs)

$$F_m = (F_{max} + F_{min})/2 \;,\; F_a = (F_{max} - F_{min})/2$$

$$\tau_a = K_w\,8\,F_a\,D/\pi\,d^3 \;,\; \tau_m = K_s\,8\,F_m\,D/\pi\,d^3$$

$$1/n = \tau_a/S_{se} + \tau_m/S_{sy} \quad \text{(Soderberg - shear)}$$

$S_{sy} = 0.577\,S_y \approx 0.45\,S_{ut}$; S_{se} (torsional endurance, 10^7 cycles) ≈ 0.2-$0.3\,S_{ut}$ (unpeened) - shot peening improves by ~ 20%. Hooks of tension springs are weak points : stress concentration ($K = (4C_1^2 - C_1 - 1)/4C_1(C_1 - 1)$).

Tension (extension) spring - special points

- Wound with *initial tension* F_i (coils pressed together) - spring starts to extend only when $F > F_i$; $F = F_i + k\delta$. Body length $L_b = (N_a + 1)\,d$; free length (with hooks) $L_f = L_b + 2\,(D - d)$ (full loop).

9. Torsion Spring

Helical wire acted on by a *moment* $M = F\,r$ about the coil axis (door hinge, clothes peg, garage door) - the wire is in *bending*, not torsion. Moment tends to close (wind up) the coils.

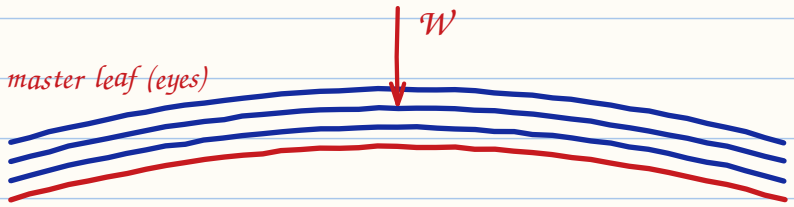
$$\sigma = K_i \cdot 32\,M / \pi\,d^3 \quad , \quad K_i = (4C^2 - C - 1)/(4C\,(C - 1)) \ \text{(inner fibre)}$$
$$\theta = 64\,M\,D\,N_a / E\,d^4 \ \text{(rad)} \quad = \quad 10.2\,M\,D\,N_a / E\,d^4 \ \text{(revolutions)}$$

- Direction of winding : load should tend to wind the spring up (increase the number of turns) → residual stress opposes working stress. Mean dia reduces as $D' = D\,N_a/(N_a + \theta_{rev})$ - leave clearance over the guide rod (≥ 10% of d).
- Stiffness (per radian) = $M/\theta = E\,d^4 / 64\,D\,N_a$; ends : straight, hinged, short or long legs ; $N_a = N_{body} + (L_1 + L_2)/(3\pi D)$ for the legs. Use $\sigma_{allow} = 0.7\,S_{ut}$ (static, cold wound, stress-relieved).

Example 6 : torsion spring $M = 3000$ N·mm, wire $d = 3$ mm, $C = 8$ ($D = 24$ mm), $E = 207$ GPa. $K_i = (256 - 8 - 1)/(4\times8\times7) = 1.10$; $\sigma = 1.10 \times 32 \times 3000/(\pi \times 27) = 1244$ MPa (within music wire's $0.7\,S_{ut}$). θ per turn $= 10.2 \times 3000 \times 24\,N_a/(207000 \times 81) \rightarrow$ for $N_a = 6$: 0.263 rev = 95°.

10. Leaf Springs

Semi-elliptic leaf spring (automobile & railway suspension) : several flat strips (leaves) of decreasing length clamped together ; acts as a beam of uniform strength. Interleaf friction gives damping.



semi-elliptic leaf spring , span 2L
each half = cantilever of length L , load P = W/2

Fig 7: Semi-elliptic multi-leaf spring

- Equivalent : a triangular plate of width n b and thickness t, cantilever length L = half span, end load P = W/2 → uniform bending stress along the length (graduated leaves). Full-length (extra) leaves n_f (master leaf with eyes) take shear, side loads ; they are over-stressed so a nip (initial gap C) is given for pre-stressing - gradually equalising stress.

Graduated leaves only : $\sigma = 6\,P\,L / n\,b\,t^2$, $\delta = 6\,P\,L^3 / E\,n\,b\,t^3$

With n_f full-length + n_g graduated leaves :

$\sigma_g = 12\,P\,L / b\,t^2\,(2n_g + 3n_f)$ $\quad \sigma_f = 18\,P\,L / b\,t^2\,(2n_g + 3n_f)$

$\delta = 12\,P\,L^3 / E\,b\,t^3\,(2n_g + 3n_f)$

- Load share : $P_g = 2\,n_g\,P / (2n_g + 3n_f)$; $P_f = 3\,n_f\,P / (2n_g + 3n_f)$; $\sigma_f = 1.5\,\sigma_g$ (hence nip). Nip $C \approx 2\,P_f\,L^3$ … chosen so that after bolting $\sigma_f = \sigma_g$. Spring rate $k = W/\delta_{total}$.
- Materials : silicon-manganese / chrome-vanadium steel (55Si2Mn90 , 50Cr1V23) ; oil hardened & tempered, shot peened ; σ_{allow} = 400-500 MPa ; leaves lubricated graphite. Fatigue : K_f at eye, centre bolt, clip - use rolled / shot peened.

Example 7 : load W = 12 kN, span 1000 mm → L = 500 , P = 6 kN. n_f = 2, n_g = 6 , b = 50 , σ_{allow} = 420 MPa. $\sigma_f = 18 \times 6000 \times 500/(50\,t^2 \times 18) = 60000/t^2 \le 420 \rightarrow t$ = 12 mm. σ_g = 278 MPa , σ_f = 417 (OK). $\delta = 12 \times 6000 \times 500^3/(207000 \times 50 \times 12^3 \times 18)$ = 28 mm (≈ 29 mm incl. camber).

Typical exam questions - Unit IV

- (1) Derive torque capacity of a disc clutch by uniform pressure and uniform wear theory ; which is used for design and why ?
- (2) Design a multi-plate clutch for given power, speed, p_{max}, μ ; find number of plates.
- (3) Explain self-energizing and self-locking of brakes ; condition for a block brake with offset fulcrum.
- (4) Derive $T_1/T_2 = e^{\mu\theta}$ and design a band brake ; differential band brake self-locking condition.
- (5) Design a helical compression spring for load F and deflection δ ; define Wahl factor, spring index.
- (6) Torsion spring stress and deflection ; design of semi-elliptic leaf spring with full-length leaves and nip.

Spring	Stress	Deflection
Helical compression / tension	$\tau = K_w\ 8FD/\pi d^3$	$\delta = 8FD^3N_a/Gd^4$
Torsion	$\sigma = K_i\ 32M/\pi d^3$	$\theta = 64\ M\ D\ N_a/E\ d^4$
Leaf (graduated)	$\sigma = 6PL/nbt^2$	$\delta = 6PL^3/Enbt^3$
Clutch (uniform wear)	$T = \mu\ W\ (r_o+r_i)/2$	$n = n_1+n_2-1$
Band brake	$T_1/T_2 = e^{\mu\theta}$	$T_b = (T_1-T_2)\ r$

UNIT - V

Design of Bearings and Gears

1. Sliding Contact (Journal) Bearings

A bearing supports a rotating / oscillating shaft and carries load while allowing relative motion with low friction. Sliding (plain) bearing : journal rubs in a sleeve with a lubricant film in between.

- Types : journal (radial load) - full 360°, partial (120°-180°), fitted ; thrust (axial) - collar, pivoted pad (Michell / Kingsbury) ; hydrodynamic (film from motion) , hydrostatic (film from external pump).
- Merits vs rolling : silent, withstands shock & high load, compact radially, cheap for large sizes, long life with film, split for crank-shaft ; Demerits : high starting friction, needs continuous oil supply, axial space ↑.

Modes (regimes) of lubrication

Mode	Film	μ	Remarks
Hydrodynamic (thick film)	full film, surfaces separated	0.001 - 0.01	needs speed, $\mu \propto Z\,N/p$
Mixed / thin film	partial asperity contact	0.01 - 0.1	start-up / low speed
Boundary	molecular oil layer only	0.05 - 0.2	oiliness, additives
Hydrostatic	pressurised oil from pump	≈ 0.001	works at zero speed
Elasto-hydrodynamic	film in highly loaded contact	0.001 - 0.005	gears, rolling bearings

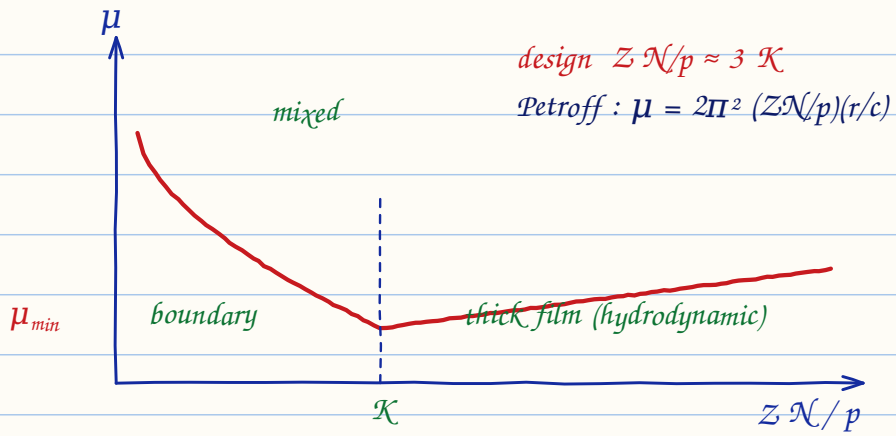


Fig 1: Stribeck curve - friction vs bearing characteristic $Z\,N/p$

- Bearing characteristic number ZN/p (Z = absolute viscosity Pa·s, N rpm, $p = W/(l\,d)$ bearing pressure). Bearing modulus K = value of ZN/p at minimum friction ; to the left of K the film breaks (mixed) → design with $ZN/p \geq 3K$ as safety, usually ZN/p well in the thick-film zone.
- Sommerfeld number $S = (ZN/p)(r/c)^2$ (dimensionless, N in rps, p in Pa) - basic design variable ; Raimondi - Boyd charts relate S to ε, h_0/c, μ r/c, flow.

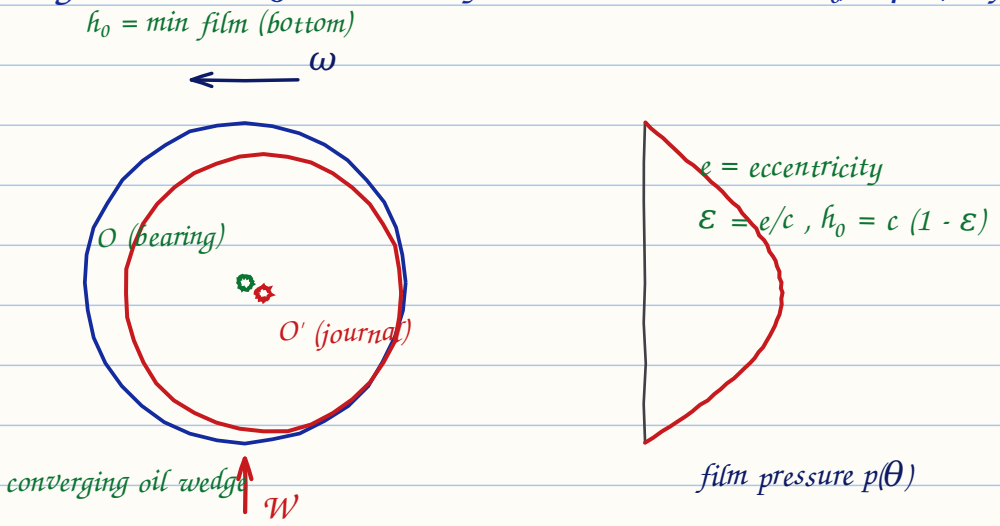


Fig 2: Hydrodynamic journal bearing - oil wedge

At rest the journal touches the bearing at the bottom. On rotation, oil is dragged into the converging wedge and a pressure builds in it that lifts the journal and shifts it sideways (attitude angle) : the shaft floats on oil. Minimum film thickness $h_0 = c\,(1 - \varepsilon)$ must exceed the surface roughness ($h_0 \geq 0.005$ mm).

Terminology

- Journal radius r , bearing radius R ; radial clearance $c = R - r$ ($c/r = 0.001$ to 0.002 or $c = 0.001\,d$) ; eccentricity e ; eccentricity ratio $\varepsilon = e/c$; l/d ratio = 0.75-2.8 (usually 1-1.5 : long → end leakage ↓, misalignment ↑) ; projected area $A = l\,d$.

Application	p (N/mm²)	l/d	Application	p (N/mm²)
Electric motor	0.7 - 1.4	1 - 2	Railway axle	3.5 - 4.5
Gear box / turbine	0.7 - 2	1 - 2	Crank pin (engine)	7 - 14
Machine tool	0.7 - 1.4	1 - 4	Main bearing (auto)	4 - 5

2. Friction in Journal Bearings - Petroff & McKee

Petroff's equation assumes the shaft is concentric (no eccentricity, no load) so the friction is just the viscous shear of the oil film :

$$\mu = 2\pi^2 (Z N / p)(r / c) \quad (N \text{ in rps , SI units})$$

$$\text{McKee : } \mu = (33 / 10^8)(Z N / p)(r / c) + k$$

- In McKee's form Z in kg/m·s (Pa·s), N in rpm, p in N/mm² ; k = correction for end leakage ≈ 0.002 for l/d = 0.75 to 2.8. The two give the straight line of the Stribeck curve ; experiments (McKee brothers) confirmed for ZN/p beyond the transition.
- Power lost in friction : $H_g = \mu W v$ (W) , $v = \pi d N / 60$ (m/s) ; torque $T_f = \mu W r$.

Heat balance (thermal equilibrium)

$$H_g = \mu W v \text{ (generated)} \le H_d = C A (t_b - t_a)$$

$$A = l d \text{ (projected area)}$$

C = heat dissipation coefficient : 140-420 W/m²·°C (unventilated), up to 1230 (well ventilated).

- Bearing temperature $t_b = t_a + (H_g / C A) \le$ 60-70°C ; oil film temperature ≈ $(t_b + t_o)/2$ where viscosity Z is taken. If $H_g > H_d \rightarrow$ increase area, use oil circulation / cooler, reduce μ.

Design procedure for a hydrodynamic journal bearing

- (1) Given W, N, d (from shaft design). (2) Choose $l/d \rightarrow l$; $p = W/(l\,d) \le p_{allow}$. (3) Select lubricant ($Z$ at operating temperature, SAE grade). (4) c/r (or c = 0.001 d). (5) Check $ZN/p \ge 3K$. (6) μ by McKee. (7) $H_g = \mu W v$; (8) $H_d = C A \Delta t \rightarrow t_b$; keep ≤ 70°C ; (9) check h_0 and ε from charts.

Example 1 : d = 50 mm, l = 75 mm (l/d = 1.5), W = 3 kN, N = 600 rpm, Z = 0.025 Pa·s, c/r = 0.001 (r/c = 1000), ambient 30°C, C = 300 W/m²°C. $p = 3000/(50 \times 75) = 0.8$ N/mm². $ZN/p = 0.025 \times 600/0.8 = 18.75$. $\mu = 33\times10^{-8} \times 18.75 \times 1000 + 0.002 =$ 0.0082. $v = \pi \times 0.05 \times 600/60 = 1.57$ m/s ; $H_g = 0.0082 \times 3000 \times 1.57 = 38.6$ W. $H_d/\Delta t = 300 \times (0.05 \times 0.075) = 1.125$ W/°C $\rightarrow \Delta t = 34$°C $\rightarrow t_b = 64$°C ≤ 70°C (OK).

Lubricants

- Absolute viscosity Z (Pa·s ; 1 cP = 10^{-3} Pa·s) ; kinematic $\nu = Z/\rho$ (cSt ; $\rho \approx 870$ kg/m³) ; viscosity falls sharply with temperature - measured by viscosity index VI (high VI = small change). SAE grades 10, 20, 30, 40, 50 (higher = thicker, for slower speed / heavier load / hotter).
- Other properties : oiliness (boundary film), flash & fire point (safety), pour point (cold start), oxidation stability, anti-wear / EP additives. Grease (oil + soap) for low speed, intermittent, difficult to feed ; solid lubricants (graphite, MoS_2) for very high temperature / vacuum.
- Supply : ring oiling, splash, bath, pressure feed through groove at the unloaded side (oil must enter where film is thick, not in the loaded zone) ; axial grooves must not run in the loaded region.

Hydrostatic & thrust bearings (short note)

- Hydrostatic : oil at pressure p_s from a pump fills pockets in the bearing → $W = p_s A_{eff}$; film exists at zero speed → no wear at start-up, very low friction, high stiffness, but pump + filter + piping cost. Used in machine tool spindles, telescopes, heavy turbines (jacking oil).
- Thrust bearing : flat collar cannot form wedge → pivoted tilting pad (Michell / Kingsbury) : pad tilts to make a wedge film ; capacity $W = p\,A$; ship propeller, turbines, vertical pumps. Step / tapered land bearings for small sizes.

3. Bearing Materials & Failures

- Requirements : compressive & fatigue strength, conformability (take misalignment), embeddability (bury dirt), low friction, thermal conductivity, corrosion & seizure resistance, cheap.

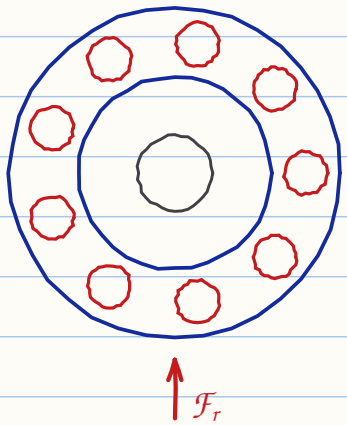
Material	Properties	Use
Tin babbitt (Sn 90, Sb 7, Cu 3)	soft, best embeddability / conformability, weak > 120°C	turbines, general
Lead babbitt	cheaper, lower strength	light duty
Copper-lead / lead bronze	high fatigue strength, to 250°C	IC engines, heavy
Phosphor / tin bronze	strong, hard shaft needed	gears, worm shafts
Cast iron	low cost, slow speeds, good with grease	machine tools
Sintered (oil-impregnated)	self-lubricating, no maintenance	fans, appliances
PTFE / nylon / carbon	dry running, corrosion proof	chemical, food

Bearing failures

- Abrasion / scoring (dirt, rough shaft) → filters, seals, hard smooth journal (Ra ≤ 0.4 µm) ; fatigue (cracks, pitting) (cyclic pressure) → thin lining ; wiping / seizure (film breakdown, overheating) → proper cooling, correct clearance ; corrosion (acid in oil) → inhibitors ; cavitation / erosion ; edge loading (misalignment) → spherical seat, relief ; fretting.

4. Rolling Contact Bearings

Load transmitted through balls / rollers between two hardened (≥ 60 HRC, 100Cr6 / SAE 52100) rings, rolling friction ≈ 0.001-0.005 : low starting friction, accurate position, standardised (ISO), interchangeable, but noisy, poor with shock, limited life (fatigue), larger radius.



heaviest ball $W_{max} \approx 5\ F_r/z$

Stribeck: $C_0 = k\, d^2\, z\, /\, 5$

$L_{10} = (C/P)^3$ million rev (balls)

$= (C/P)^{10/3}$ (rollers)

$P = X\, V\, F_r + Y\, F_a$

Fig 3: Deep groove ball bearing & load relations

Type	Load	Remarks
Deep groove ball (6xxx)	radial + some axial both ways	most common, high speed
Angular contact (7xxx)	radial + large axial (one way)	paired / preloaded, spindles
Self-aligning ball (1xxx)	radial, tolerate 2-3° misalignment	line shafts
Cylindrical roller (NU, N)	heavy radial, no axial (NJ small)	motors, gearboxes
Tapered roller (3xxx)	heavy radial + axial one way	wheel hubs, pairs
Spherical roller (2xxx)	very heavy + misalignment	mining, rolls
Thrust ball / roller (5xxx)	axial only	pivots, vertical shafts

Bearing designation (ISO)

e.g. 6205 - 2RS : 6 = deep groove ball ; 2 = light series (dimension / width) ; 05 → bore = 05 × 5 = 25 mm (04 → 20 mm ; for bore 20-480 mm : last two digits × 5 ; 00 → 10 , 01 → 12 , 02 → 15 , 03 → 17 mm). Suffix : Z shield , 2RS rubber seals , C3 larger clearance.

- Fits : rotating ring (inner for rotating shaft) - interference fit (k5 / m5) on shaft ; stationary ring - transition / slip fit in housing (H7) to allow creep ; shoulders ≥ fillet radius of bearing ; axial location with circlips / nut.
- Failures : subsurface fatigue spalling (normal life end), brinelling (static / shock overload), false brinelling, wear (dirt), corrosion, cage break, smearing (skid), electrical fluting. Lubrication : grease (≤ 0.3-0.5 × limiting speed, sealed-for-life) or oil bath / mist (high speed).

5. Static & Dynamic Load Capacity

Static load capacity C_0 - Stribeck's equation

Static capacity = radial load for which the permanent deformation at the most heavily loaded rolling element contact = 0.0001 × diameter of rolling element (does not impair smooth running). Stribeck : the max load on a ball $W_{max} = 5\ W/z$ (clearance) ≈ $4.37\ W/z$ (no clearance) ; W_{max} limited by Hertz contact stress → $W_{max} = k\ d^2$ →

$$C_0 = k\ d^2\ z\ /\ 5 \quad \text{(ball)} \quad ; \quad C_0 = k\ l\ d\ z\ /\ 5 \quad \text{(cylindrical roller)}$$

static safety : $C_0 \geq s_0\ P_0$ (s_0 = 1 normal , 1.5-2 for shock / quiet)

k = constant depending on materials and radii of curvature of contact (N/mm^2) ; d = ball dia ; z = number of balls ; l = roller length. Equivalent static load $P_0 = 0.6\ F_r + 0.5\ F_a$ ($\geq F_r$) for deep groove ball bearing.

Dynamic load capacity C & L_{10} life

- Basic dynamic capacity C = constant radial load for which 90% of a group of identical bearings complete 10^6 revolutions without fatigue failure (rating life L_{10}). Catalogue gives C and C_0.

$$L_{10} = (C / P)^k \text{ million rev} ; k = 3 \text{ (ball)}, 10/3 \text{ (roller)}$$
$$L_{10h} = 10^6\ (C/P)^k / 60\ N \quad \text{(hours)}$$
$$C_{required} = P\ (L_{10})^{1/k} \ , \ L_{10} = 60\ N\ L_{10h}/10^6$$

- Load-life : doubling P cuts the ball bearing life to 1/8 (cubic law). Reliability other than 90% : $L_R = a_1\ L_{10}$; a_1 = 0.62 (95%), 0.53 (96%), 0.44 (97%), 0.33 (98%), 0.21 (99%).
- Variable loads / speeds : $P_{eq} = (\Sigma\ P_i^3\ n_i\ /\ \Sigma\ n_i)^{1/3}$; equivalent speed $N_{eq} = \Sigma\ N_i\ t_i$; Palmgren-Miner : $\Sigma\ (n_i/L_i) = 1/L$.

Machine	L_{10h} (h)
Household / agricultural / instruments	300 - 3000
Light electric motors, fans (8 h/day)	8000 - 15000
Gear boxes, machine tools (8 h/day)	20000 - 30000
Pumps, compressors (24 h/day)	40000 - 60000
Power-plant, mines - continuous	100000+

Equivalent bearing load

$$P = K_a\ (X\ V\ F_r + Y\ F_a)$$

- V = 1 (inner ring rotates), 1.2 (outer ring rotates) ; load (service) factor K_a : steady 1.0-1.1 ; light shock 1.2-1.5 ; moderate shock 1.5-2 ; heavy shock 2-3. X, Y from catalogue : if $F_a/(V\ F_r) \leq e \rightarrow X = 1$, $Y = 0$; if $> e \rightarrow X = 0.56$ and Y from table (deep groove) :

F_a/C_0	0.025	0.04	0.07	0.13	0.25	0.5
e	0.22	0.24	0.27	0.31	0.37	0.44
Y	2.0	1.8	1.6	1.4	1.2	1.0

Example 2a (variable load) : a ball bearing (C = 25 kN) runs at 800 rpm : 5 kN for 40% of time, 3 kN for 35%, 1.5 kN for 25%. $P_{eq} = (5^3 \times 0.4 + 3^3 \times 0.35 + 1.5^3 \times 0.25)^{1/3} = (60.3)^{1/3} = 3.92$ *kN.* $L_{10} = (25/3.92)^3 = 259$ *million rev →* $L_{10h} = 259\times10^6/(60 \times 800) = 5400$ *h.*

6. Selection of Bearings from Catalogue

- *(1) Radial* F_r*, axial* F_a*, speed* N*, shaft dia d (bore), required* L_{10h}*, reliability, environment. (2) Choose type (ball for light / high speed, roller for heavy, tapered for combined, thrust for pure axial, self-aligning if misaligned). (3)* $L_{10} = 60\ N\ L_{10h}/10^6$*. (4)* $P = K_a(X\ V\ F_r + Y\ F_a)$ *(guess* C_0 *first, iterate). (5)* $C_{req} = P\ L_{10}^{1/k}$*. (6) Select bearing with bore d and* $C \geq C_{req}$*. (7) Check* $C_0 \geq s_0\ P_0$ *and limiting speed.*

Example 2 : shaft d = 40 mm, $F_r = 4$ *kN,* $F_a = 1.5$ *kN,* $N = 1000$ *rpm,* $L_{10h} = 5000$ *h, inner ring rotates, moderate shock* $K_a = 1.2$*.* $L_{10} = 60 \times 1000 \times 5000/10^6 = 300$ *million rev. Try 6308 (typical C = 41 kN,* $C_0 = 24$ *kN) :* $F_a/C_0 = 0.0625 \rightarrow e \approx 0.26$ *;* $F_a/F_r = 0.375 > e \rightarrow X = 0.56$*,* $Y \approx 1.7$*.* $P = 1.2\ (0.56 \times 4 + 1.7 \times 1.5) = 5.75$ *kN.* $C_{req} = 5.75 \times 300^{1/3} = 5.75 \times 6.69 = 38.5$ *kN ≤ 41 (OK) → 6308 is safe (check against actual catalogue values).*

7. Spur Gears - Terminology & Tooth Forces

Spur gear : straight teeth parallel to axis, transmits power between parallel shafts.

Preferred module series (mm) : 1 , 1.25 , 1.5 , 2 , 2.5 , 3 , 4 , 5 , 6 , 8 , 10 , 12 , 16 , 20 .

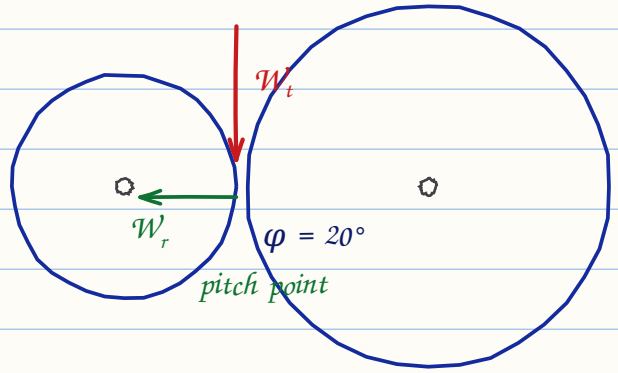
$$\text{module } m = d / z \quad ; \quad \text{circular pitch } p_c = \pi\ m \quad ; \quad d = m\ z$$

$$\text{addendum} = m \ ; \ \text{dedendum} = 1.25\ m \ ; \ \text{clearance } 0.25\ m \ ; \ \text{tooth depth } 2.25\ m$$

$$i = N_1/N_2 = z_2/z_1 \quad ; \quad \text{centre distance } a = m\ (z_1 + z_2)/2$$

- *Pressure angle* $\varphi = 20°$ *(full-depth involute) , 14.5° , 20° stub ; minimum teeth to avoid interference* $z_{min} = 2 / \sin^2\varphi = 17\ (20°), 32\ (14.5°)$ *; face width b = 9.5-12.5 m (usually 10 m) ;* $z_{pinion} \geq 18$*-20 for steel.*

- Materials : cast iron FG 200-260 ($\sigma_b \approx$ 50-70 MPa) ; C45 / 40C8 (σ_b 140-200) ; alloy steel 40Ni14, 35Mn2Mo28 ; case hardened 20MnCr5 (BHN 600) ; cast steel CS 540 ; phosphor bronze (worm wheel) ; nylon / fibre (silent).



$W_t = 2T/d = P/v$

$W_r = W_t \tan \varphi$

$W_n = W_t / \cos \varphi$

$v = \pi d N / 60$

Fig 4: Forces on a spur gear tooth at the pitch point

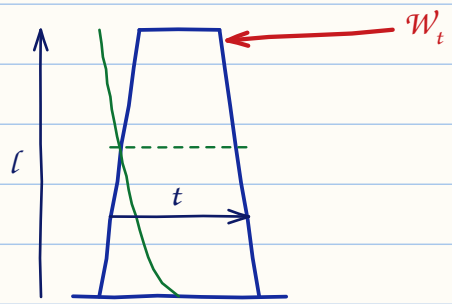
- Failure modes : tooth breakage (bending fatigue at root - beam strength), pitting (surface contact fatigue - wear strength), scoring (scuffing - lubrication), abrasive wear, plastic flow (soft gears).

Involute tooth profile - key facts

- Involute = curve traced by a point on a taut string unwound from the base circle ($d_b = d \cos \varphi$). Conjugate action : constant velocity ratio even if centre distance changes slightly ; line of action tangent to both base circles passes through the pitch point at angle φ ; teeth contact along the line of action only.
- Contact ratio = length of path of contact / base pitch ≥ 1.2 (at least 1.4 desirable) - average number of teeth in mesh ; 20° gears ≈ 1.6-1.8. Smooth, quiet, less load per tooth if > 1.
- Interference : tip of gear tooth cuts the root of pinion below the base circle → weak tooth. Avoided by $z \geq z_{min}$ (17 for 20°), stub teeth, increasing φ, or profile shift (corrected gears). Undercutting in hobbing small pinions weakens the root.
- Cycloidal profile : no interference but needs exact centre distance, costly - used in watches. Gear quality (error e) by class : precision ground (e ≈ 0.01 mm) → commercial cut (0.1 mm).

8. Beam Strength - Lewis Equation

Lewis (1892) : tooth is a cantilever beam of uniform strength (a parabola inscribed in the tooth profile) loaded by the tangential component W_t at the tip (assumes full load on one tooth, no impact).



tooth = cantilever, critical section t at root

$\sigma_b = 6\, W_t\, l / (b\, t^2)$

$W_t = \sigma_b\, b\, m\, Y$ (Lewis)

$Y = \pi\, y = 0.484 - 2.87/z$ (20° full depth)

Fig 5: Lewis model of a gear tooth

$$\sigma_b = 6\, W_t\, l / b\, t^2 = W_t / (b\, m\, Y)$$

Beam strength $S_b = \sigma_b\, b\, m\, Y$ (N) ≥ effective load F_{eff}

Tooth system	Form factor y (Lewis)	$Y = \pi\, y$
20° full depth	$y = 0.154 - 0.912/z$	$0.484 - 2.87/z$
14.5° composite / full depth	$y = 0.124 - 0.684/z$	$0.390 - 2.15/z$
20° stub	$y = 0.175 - 0.841/z$	$0.55 - 2.64/z$

- Pinion vs gear : the weaker of the pair is the one with smaller $\sigma_b\, Y$ (Y smaller for fewer teeth) ; design the pinion if $\sigma_{b1}\, Y_1 < \sigma_{b2}\, Y_2$, else the gear. Same material → pinion is weaker.
- σ_b allowable = $S_{ut}/3$ (static design stress, steel) ; endurance based (fatigue) lower ; typical : C45 normalised 140 MPa, hardened 200, FG 200 : 55.

9. Design for Dynamic Load & Wear Load

Dynamic (velocity) factor

$$F_t = P / v \quad ; \quad v = \pi\, d_1\, N_1 / 60 \ \text{(m/s)}$$

$= 3 / (3 + v)$ ($v < 7.5$ m/s , ordinary cut) ; $6/(6 + v)$ (v up to 12.5 , carefully cut) ; $5.55/(5.55 + \sqrt{}$

Effective load (Barth / Lewis) : $F_{eff} = C_s\, F_t / C_v \le S_b$

Service factor C_s : uniform load 1.0-1.25 ; light shock 1.25-1.5 ; moderate 1.5-1.75 ; heavy 1.75-2.25.

Buckingham dynamic load

$$F_d = F_t + F_i \quad , \quad F_i = 21\ v\ (b\ C + F_t) / (21\ v + \sqrt{(b\ C + F_t)})$$

Design : $S_b \geq F_d$ (with 1.25 - 1.5 factor of safety)

F_i = incremental load from tooth errors e (mm) and elasticity ; C = deformation factor = e / (1/E_1·...) listed for materials, e.g. error e = 0.05 mm steel-steel 20° → C ≈ 700 N/mm. v in m/s, b in mm. Higher accuracy gear → smaller e → smaller F_i.

Wear (surface) strength - Buckingham

$$S_w = b\ Q\ d_1\ K \quad \geq \quad F_d \quad (\text{or } F_{eff})$$

$Q = 2\ z_2 / (z_1 + z_2) = 2i/(i + 1)$ (external gears) ; $K = 0.16\ (BHN/100)^2$ N/mm² (20

- K = load-stress factor = $\sigma_{es}^2 \sin\varphi\ (1/E_1 + 1/E_2) / 1.4$ where σ_{es} = surface endurance limit = 2.75 BHN - 70 (N/mm²). Use the lower BHN of the pair. Q = $2z_2/(z_2 - z_1)$ for internal gears.

Spur gear design procedure

- (1) P, N_1, i. (2) $z_1 \approx 20$, $z_2 = i\ z_1$; select materials → σ_{b1}, σ_{b2}. (3) Y_1, Y_2 → weaker gear. (4) $b = 10\ m$; $v = \pi\ m\ z_1\ N_1/60000$; $F_t = P/v$; C_v. (5) $S_b = F_{eff}$ → solve m ; take standard module. (6) d_1, d_2, b, a. (7) Check dynamic load $F_d \leq S_b$ and wear $S_w \geq F_d$. (8) Check lubrication & hardness ; sketch with dimensions.

Example 3 : 15 kW, N_1 = 1000 rpm, i = 3, z_1 = 20 , z_2 = 60 , 20° full depth, σ_{b1} = 180 (pinion) , σ_{b2} = 120 MPa (gear), BHN 250, C_s = 1.25. Y_1 = 0.484 - 2.87/20 = 0.341 ; Y_2 = 0.436 → σY : pinion 61.4 ; gear 52.3 → gear weaker.

$b = 10\ m$; $v = \pi\ m \times 20 \times 1000/60000 = 1.047\ m$; $F_t = 14324/m$; $C_v = 3/(3 + 1.047\ m)$. $S_b = 120 \times 10m \times m \times 0.436 = 523\ m^2$. Equate : $523\ m^2 = 1.25 \times (14324/m)(3 + 1.047\ m)/3$ → $m^3 - 11.9\ m - 34.2 = 0$ → m = 4.4 → m = 5 mm (standard).

d_1 = 100 , d_2 = 300 , b = 50 mm , a = 200 mm. v = 5.24 m/s ; F_t = 2865 N ; C_v = 0.364 ; $F_{eff} = 1.25 \times 2865/0.364 = 9840$ N < $S_b = 120 \times 50 \times 5 \times 0.436 = 13090$ N (OK).

Wear : $Q = 2 \times 60/80 = 1.5$; $K = 0.16\ (2.5)^2 = 1.0$ → $S_w = 50 \times 1.5 \times 100 \times 1.0 = 7500$ N < 9840 (not OK) → raise hardness to BHN 350 : K = 1.96 → S_w = 14700 N (OK) (surface hardened).

Example 4 (trial $m = 4$) : same drive (15 kW, $N_1 = 1000$, $z_1 = 20$, $i = 3$), both gears hardened to BHN 400 → $K = 0.16 \times 4^2 = 2.56$ N/mm². $m = 4$: $d_1 = 80$, $b = 40$, $v = 4.19$ m/s , $F_t = 3581$ N , $C_v = 0.417$ → $F_{eff} = 1.25 \times 3581/0.417 = 10730$ N.

Wear : $S_w = b\,Q\,d_1\,K = 40 \times 1.5 \times 80 \times 2.56 = 12290$ N ≥ 10730 (OK). Bending (gear, $\sigma_b = 120$) : $S_b = 120 \times 40 \times 4 \times 0.436 = 8370$ N < 10730 (not OK) → bending governs, so take $m = 5$ as in Example 3.

- Lesson : check both beam strength and wear strength - hardness helps wear ($K \propto BHN^2$) but not bending (depends on σ_b and Y) ; the larger module required decides.

Typical exam questions - Unit V

- (1) Explain the modes of lubrication & Stribeck curve ; define bearing modulus. (2) Petroff's equation and McKee's equation ; heat balance. (3) Design a journal bearing for given W, N, d.
- (4) Stribeck's equation for static load ; L_{10} life & load-life relation ; select a ball bearing from catalogue. (5) Lewis equation, derivation, Buckingham dynamic and wear loads ; design a pair of spur gears.

Topic	Key formula
Petroff	$\mu = 2\pi^2 (ZN/p)(r/c)$
Heat balance	$\mu\, W\, v = C\, A\, (t_b - t_a)$
Stribeck	$C_0 = k\, d^2\, z\, /\, 5$
Bearing life	$L_{10} = (C/P)^3$ (ball)
Lewis	$W_t = \sigma_b\, b\, m\, Y$
Wear	$S_w = b\, Q\, d_1\, K$